Sequential Validation of Investment Policies: Trading Costs, Moment Information, and the Cost of Waiting

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Abstract

Sequential validation delays a portfolio change while the current policy continues to trade. We study the moment information needed to certify a quadratic-utility improvement and the utility forgone while waiting. Two costless cases isolate the mechanisms. Over an iid fixed-policy return class, a public second-moment cap gives an exact ENTRY testing frontier: informative uniformly valid testing requires mean return above half risk aversion times that cap. Cash-PARK has an economic score lower bound. For a stationary Gaussian estimated policy, information grows only logarithmically; an explicit adaptive null gives a matching high-confidence exponent and an attained pathwise coefficient for conditional and realized cumulative utility forgone. These are not expected stopped-cost results. With finitely many signals and positive ridge, estimating the full matrix of raw second moments under a leverage constraint gives the same coefficient. In that instance, a constant certificate makes maximal power vanish in the high-confidence limit uniformly over predictable Catoni bets. For portfolios with trading costs, observable comparisons retain separate allocation states and public conditional moment certificates. Under constant effects we distinguish estimating a quadratic-certificate bet, optimizing log growth and targeting deadline power. Controlled experiments and financial pilot paths separate validity, feasible certification and economic performance. A separately frozen stronger-moment extension allows entry but can require substantial waiting. The financial tests have individual guarantees; rejection certifies neither current profitability nor search-wide safety. Empirical applications are outside the current scope.

Keywords: sequential inference; investment signals; e-processes; transaction costs; portfolio estimation; simulation.

1 Introduction

A portfolio manager faces three related decisions: whether to add a signal, park an incumbent signal, or replace one signal with another. The economically relevant object is the change in a feasible investment policy. A predictor's own coefficient, its standalone performance and its contribution to a portfolio are different quantities. A signal without a direct predictive coefficient may hedge an incumbent exposure or proxy another characteristic. Conversely, an authentic predictor may worsen a portfolio when estimated weights and trading costs are taken into account.

Sequential validation takes time while the incumbent policy continues to trade. We study the information required to certify a utility improvement and the utility forgone before certification. A valid evidence process and a profitable trading rule address different parts of this problem.

We formulate the decisions as comparisons of two policies constructed before the returns used to evaluate them. Each policy pays its own trading costs and keeps its own allocation state. Their observable utility difference feeds a heavy-tail-robust e-process. An annual search

proposes additions, removals and a restricted set of swaps; the test evaluates those proposals. This separation allows one to ask both whether the statistical comparison is valid and whether waiting for its evidence is worthwhile. We call the annual add–remove–swap implementation PARK/REVIVE; its name describes available actions, not a guarantee of adaptation to a new regime.

Two explicit cases isolate the information and waiting requirements. Section 13 characterizes fixed-policy ENTRY over all iid return laws with a public second-moment cap. Positive utility alone need not permit informative uniformly valid testing; we give the exact frontier and a sufficient higher-moment route to consistency. Cash-PARK instead has a one-sided economic score bound. Section 14 treats a stationary Gaussian estimated reference whose cash disadvantage shrinks over time. Its cumulative information grows as $\log n$. An admissible adaptive null gives a matching confidence exponent. With risk aversion γ, the attained coefficient of conditional and realized utility forgone is $4/\gamma$ per $\log(1/\alpha)$ at test level α. This coefficient is a high-confidence limit for that instance, not an expected stopped cost or a claim about the complete financial strategy. An extension with finitely many signals, positive ridge and an estimated second-moment matrix retains the same stopped-utility coefficient; Section 15 computes how the time exponent changes with the learner's limiting curvature. For the same policy, a constant-certificate Catoni family has vanishing maximal high-confidence power over all predictable bets, including rules chosen separately at each confidence level. The fixed penalty constrains that whole family despite growing information in the return experiment. With a geometry-dependent certificate, a past-only raw-moment bet attains the best asymptotic log rate among fixed bounded predictable rules. An oversized public second-moment cap still increases its attained waiting coefficient; Proposition 19 quantifies that loss.

The supporting results connect utility comparisons to implementable portfolios. T1 establishes individual episode validity. T2 bounds stationary power and the estimation loss in a quadratic growth certificate. T3 describes population entry value under trading costs; T4 controls objective estimation and trading-state errors. Public policy geometry and corrected tail betting then provide a separately declared stronger-moment implementation. The controlled study in Section 6 distinguishes certificate, log-growth and deadline-power objectives; its predictable Bellman policy is not substituted into the financial experiment. Section 17 assesses actual fund costs and waiting, separately from certification.

The statistical ingredients build on Catoni-style confidence sequences [1] and the broader theory of safe anytime-valid inference [2]. Sequential comparison of forecast scores is developed by [7] and [8]; the latter also treats a weak null about average conditional score differences. Our T1 uses the datewise conditional-mean null stated below. It does not establish validity for that broader weak null. All sleeve returns are observed, so the present setting permits full-information counterfactual score calculation rather than requiring off-policy importance weights. We specialize these comparisons to estimated constrained portfolios, separate trading states and explicit moment certificates. Sequential scoring-rule selection with exponentiated score products is also studied in [19], under its stated conditional-factor and moment assumptions. Sequential selection between estimated forecasting methods is therefore an existing research problem. The economic optimization uses the convex trading perspective of [3]. The present contribution is the explicit connection of these components to comparisons of estimated investment policies with separately accounted trading states. We do not claim a new general e-process construction or a new general Schur-complement identity.

Portfolio evaluation of characteristics with risk and trading costs already appears in [9]; its stock-level trading-netting mechanism is outside our sleeve-turnover cost model. [10] develops GMM inference for transaction-cost effects under recursive utility. E-process applications in finance include risk-forecast backtesting [11] and comparative risk-measure backtests [12]. These precedents preclude a claim that either economic signal evaluation or sequential financial comparison is new. The contribution studied here is their specialization to observable

policy-pair scores with separate allocation states, explicit public moment certificates and an assessment of the cost of waiting. General instance log optimality and rejection-time theory for portfolio-based e-processes are developed in [13]; the portfolios there allocate statistical betting capital. Our investible portfolios generate utility scores, and T2 below is only a bound on a quadratic growth certificate, not an instance optimality result.

Horizon-aware betting and direct optimization of rejection-time objectives are already developed in [22, 23, 24]. Our controlled Catoni calculation specializes that approach; it introduces no general deadline-power principle and does not benchmark their learned or clinical-trial designs.

Estimation noise in comparisons of nested forecasting models is already analyzed by [15, 16]. In particular, fitting redundant parameters can worsen an estimated forecast even when the population models coincide. We retain such estimation effects when they change traded utility: our inferential target is an implemented policy comparison, rather than an estimation-noise-adjusted test of population predictive content. Portfolio utility losses from estimating means and covariance matrices are already analyzed by [30]. Recent parametric work derives leading minimax and Bayes expected-regret constants for portfolio choice [31]. Our stationary calculation fixes the reference learner and studies stopping under a sequential evidence constraint. Its stopped cumulative-utility coefficient is a different objective from optimizing the allocation rule's expected regret.

Costly learning and delayed adoption have established economic treatments. [27] optimizes discounted project value with an explicit non-adoption penalty; [28] combines e-value approval with subsidized experimentation and belief-state control. Our cost calculation fixes a frequentist evidence constraint and allows the implemented reference to evolve. General links between log growth and threshold-time optimality, including distinctions between convergence modes, are studied in [29]. The Gaussian instance here has a logarithmic information clock and a potentially decreasing capital; its normalization is proved directly.

Multiple testing remains a separate issue. The financial significance of searching among many predictors is emphasized by [4]. Sequential model confidence sets already provide simultaneous time-uniform coverage for superior forecasters [18]. Our specified algorithm deliberately applies no correction across candidates or episodes; its individual 5% guarantee is therefore not a 5% guarantee for the entire search. Nor does an episode rejection certify that a previously profitable signal remains profitable in the current regime.

An explicit counterexample below makes this distinction operational: an annual decision can pass both the evidence threshold and the positive-history filter while the current gain is negative.

2 The economic object

Let $A_{t+1} \in \mathbb{R}^p$ be next-period gross returns of observable signal sleeves. Investor information $\mathcal{G}_t$ contains observed characteristics and returns, published risk caps, policy states and past algorithmic decisions, but no supplied hidden market state or future shock. Allocations w_t, previous targets b_t and bets are $\mathcal{G}_t$-measurable. In the mathematical results, $\mathcal{F}_t \supseteq \mathcal{G}_t$ denotes the specified pre-return conditioning filtration; it may equal $\mathcal{G}_t$. The simulation evaluator uses an enlarged filtration also containing current latent coefficients and the current regime,

but no future innovations. Its state-conditioned oracle moments need not equal the learner's conditional moments given $\mathcal{G}_t$. Define

$$C_t(w,b) = c\|w-b\|_1 + \frac{\eta}{2}(w-b)'\Lambda_t(w-b), \tag{1}$$

$$U_{t+1}(w;b) = w'A_{t+1} - \frac{\gamma}{2}(w'A_{t+1})^2 - C_t(w,b), \tag{2}$$

$$J_t(w;b) = w'\mu_t - \frac{\gamma}{2}w'M_t w - C_t(w,b), \tag{3}$$

where $\mu_t = \mathbb{E}[A_{t+1} \mid \mathcal{F}_t]$ and $M_t = \mathbb{E}[A_{t+1}A_{t+1}' \mid \mathcal{F}_t]$. We require $\gamma > 0$, $c, \eta \geq 0$ and $\Lambda_t \succeq 0$. Whenever the expectations exist,

$$\mathbb{E}[U_{t+1}(w;b) \mid \mathcal{F}_t] = J_t(w;b). \tag{4}$$

This follows by conditional linearity and the raw-second-moment identity. The conditional null and moment assumptions below must hold for the same chosen filtration. The analytic public envelopes hold also for the enlarged simulation filtration and therefore, by the tower property, for the investor filtration. Economic event labels use the former; they do not establish the truth or falsity of the investor-filtration episode null.

The raw second moment is essential: $M_t = \Sigma_t + \mu_t\mu_t'$. Replacing M_t by covariance removes the term $\gamma(w'\mu_t)^2/2$ and changes the objective. Trading costs are subtracted after applying the quadratic score to gross returns. Squaring net returns would also define another objective. Quadratic utility is a scoring criterion; it is not globally increasing over unbounded returns.

For a support $S \subseteq [p]$, let

$$\mathcal{W}(S) = \{w : \mathrm{supp}(w) \subseteq S,\ \|w\|_1 \leq L\}, \qquad V_t(S;b) = \max_{w \in \mathcal{W}(S)} J_t(w;b). \tag{5}$$

The empty support permits only $w = 0$. Thus $V_t(\varnothing;0) = 0$, whereas $V_t(\varnothing;b) = -C_t(0,b)$: liquidation is not free. For $S \subseteq B$, feasible-set inclusion gives $V_t(B;b) \geq V_t(S;b)$ at the same state. This does not imply that an estimated policy with additional signals improves realized or conditional utility.

The implementation uses $b_t = w_{t-1}$, previous target allocations across sleeves. This stylized allocator-turnover model does not reconstruct trades inside each sleeve. Its net return for wealth and Sharpe calculations is $r_{t+1}^{net} = w_t'A_{t+1} - C_t(w_t, b_t)$; its utility remains (2). Daily returns are decimals, c is utility per unit of gross allocation change, $\eta\Lambda_t$ is quadratic cost curvature and γ is quadratic return curvature.

3 Comparison episodes and the information protocol

An episode is identified by its start τ, reference support S, candidate support B, learning rule, cost parameters and state convention. A forward move sets $B = S \cup \{j\}$; a removal sets $B = S \setminus \{j\}$; a swap sets $B = (S \setminus \{i\}) \cup \{j\}$. All use the same orientation,

$$Y_{t+1} = U_{t+1}(B) - U_{t+1}(S). \tag{6}$$

The removal candidate is the policy without the incumbent signal.

At episode inception, the two shadow policies start from the fund's same actual allocation. Thereafter, each shadow keeps its own previous target and pays its own turnover and impact. Selecting a candidate changes the fund's support; its next target is recomputed from the fund's actual previous allocation. The shadow's accumulated positions are not transferred to the fund.

Estimation uses expanding observations and is refreshed annually. Evidence uses only utility scores realized after the episode starts. A routine refit of a predeclared adaptive policy

does not rewrite its old scores. Changing the reference support closes its comparisons and opens new episodes, initialized at one, on future observations. Expanding estimation and accumulating test evidence are distinct operations.

4 Individual sequential validity

We test the episode null

$$H_0 : \quad \mathbb{E}[Y_t \mid \mathcal{F}_{t-1}] \le 0 \quad \text{at every episode date.} \tag{7}$$

This is stronger than a statement about the unconditional historical average. Let $v_t^2 > 0$ be predictable with $\mathbb{E}[Y_t^2 \mid \mathcal{F}_{t-1}] \le v_t^2$. Define the odd Catoni function

$$\phi(x) = \operatorname{sign}(x) \log(1 + |x| + x^2/2). \tag{8}$$

It satisfies, for all real x,

$$e^{\phi(x)} \le 1 + x + x^2/2, \quad e^{-\phi(x)} \le 1 - x + x^2/2, \quad \phi(x) \ge x - x^2/2. \tag{9}$$

Indeed $(1 + x + x^2/2)(1 - x + x^2/2) = 1 + x^4/4 \ge 1$; the first two inequalities follow by considering the sign of x. The last follows from $\phi(x) \ge -\log(1 - x + x^2/2) \ge x - x^2/2$.

For predictable $\lambda_t \ge 0$, set

$$L_{\tau,t} = \sum_{s=\tau+1}^{t} \{\phi(\lambda_s Y_s) - \lambda_s^2 v_s^2/2\}, \qquad E_{\tau,t} = \exp(L_{\tau,t}). \tag{10}$$

The implementation stores L and compares it with $\log(1/\alpha)$.

Theorem 1 (T1: conditional episode validity)**.** *Let τ be an almost surely finite stopping time and let the comparison rule be chosen using $\mathcal{F}_\tau$. Under (7) and the stated conditional second-moment bound, $E_{\tau,t}$ is a nonnegative supermartingale starting at one. For $0 < \alpha < 1$,*

$$\mathbb{P}\left(\sup_{t \ge \tau} E_{\tau,t} \ge 1/\alpha \mid \mathcal{F}_\tau \right) \le \alpha. \tag{11}$$

The bound also holds when decisions are restricted to annual revisions.

Proof. For $Q_t = \exp\{\phi(\lambda_t Y_t) - \lambda_t^2 v_t^2/2\}$,

$$\begin{aligned} \mathbb{E}[Q_t \mid \mathcal{F}_{t-1}] &\le e^{-\lambda_t^2 v_t^2/2}\{1 + \lambda_t \mathbb{E}[Y_t \mid \mathcal{F}_{t-1}] + \lambda_t^2 \mathbb{E}[Y_t^2 \mid \mathcal{F}_{t-1}]/2\} \\ &\le e^{-\lambda_t^2 v_t^2/2}(1 + \lambda_t^2 v_t^2/2) \le 1. \end{aligned}$$

Multiplying by past capital proves the supermartingale property. Stop at the first threshold crossing or a finite horizon. Conditional expectation of stopped capital is at most one, and crossing requires capital at least $1/\alpha$. Conditional Markov and then monotone convergence of the crossing events prove the result. Annual crossings form a subset of all crossings. □

The theorem permits past-based candidate choice, annual refitting and estimated bets. It does not combine validity across different comparisons or restarts. A useful anytime p-value for the same episode is $\min\{1, 1/\max_{s \le t} E_{\tau,s}\}$. Thus the relevant contrast with the benchmark is sequentially valid inference versus repeated uncorrected fixed-time inference, rather than e-values versus every possible p-value procedure. The operational decision uses current annual evidence, not a stale historical maximum.

Proposition 2 (Episode evidence does not certify the current sign)**.** *Even with a valid moment bound and positive episode mean, an annual rejection can occur when the current conditional gain is negative.*

Proof. Consider a deterministic 252-date episode, with $Y_t = 1/2$ for the first 200 dates and $Y_t = -1/2$ for the remaining 52. The valid second-moment bound $\nu^2 = 1$ and a predeclared first-block bet $\lambda = .1$ give $L_{252} = 148\phi(.05) - 252(.1)^2/2 > 6 > \log 20$. The episode mean is $74/252 > 0$, but $\mathbb{E}[Y_{252} \mid \mathcal{F}_{251}] = -1/2$. The episode null is false, so T1 is not violated; the interpretation as a certificate of current benefit fails. □

The issue is more fundamental for prospective certification without any restriction relating the past to the next period.

Proposition 3 (A limit under unrestricted nonstationarity)**.** *Fix a decision date t and a class of laws closed under replacing the next score by zero while preserving the law of $\mathcal{F}_t$. Let a decision $D_t \in \{0, 1\}$ be $\mathcal{F}_t$-measurable. If $\mathbb{P}_Q(D_t = 1) \leq \alpha$ for every law Q in the class satisfying $\mathbb{E}_Q[Y_{t+1} \mid \mathcal{F}_t] \leq 0$, then $\mathbb{P}_P(D_t = 1) \leq \alpha$ for every law P in the class, including alternatives with positive next-period gain.*

Proof. For any P, construct Q with the same distribution on $\mathcal{F}_t$ and $Y_{t+1} = 0$. This replacement respects any positive upper second-moment bound. The prospective null holds under Q. Since D_t depends only on the shared history, $\mathbb{P}_P(D_t = 1) = \mathbb{P}_Q(D_t = 1) \leq \alpha$. □

The closure condition deliberately permits unrestricted changes; it need not hold in the specified persistent DGP. The proposition identifies why a uniformly informative certificate of next-period benefit requires temporal structure or a different inferential target. It is not a failure of the episode-wide guarantee in T1, and no novelty claim is attached to this basic observational-equivalence argument.

5 Power and predictable betting

Work in local episode time with blocks of h observations and constant bound ν^2. For the fixed-bet result, assume

$$\mathbb{E}[Y_t \mid \mathcal{F}_{t-1}] \geq \delta > 0, \qquad \mathbb{E}[Y_t^2 \mid \mathcal{F}_{t-1}] \leq \nu^2. \tag{12}$$

Write $g(\lambda) = \lambda\delta - \lambda^2\nu^2$ and $\pi_K = \mathbb{P}(\max_{k \leq K} E_{hk} \geq 1/\alpha)$.

Theorem 4 (T2a: a fixed-bet power certificate)**.** *For any constant $\lambda \geq 0$,*

$$\pi_K(\lambda) \geq [1 - \alpha^{-1} e^{-hKg(\lambda)}]_+. \tag{13}$$

The exponent is maximized at $\lambda^* = \delta/(2\nu^2)$ with $g^* = \delta^2/(4\nu^2)$. A sufficient horizon for power at least $1 - \beta$ is

$$hK \geq \frac{4\nu^2}{\delta^2} \log \frac{1}{\alpha\beta}. \tag{14}$$

Proof. By (9) and (12),

$$\begin{aligned} \mathbb{E}[e^{-\phi(\lambda Y_t) + \lambda^2\nu^2/2} \mid \mathcal{F}_{t-1}] &\leq e^{\lambda^2\nu^2/2}(1 - \lambda\delta + \lambda^2\nu^2/2) \\ &\leq e^{-\lambda\delta + \lambda^2\nu^2} = e^{-g(\lambda)}. \end{aligned}$$

Iteration gives $\mathbb{E}E_{hK}^{-1} \leq e^{-hKg(\lambda)}$. No annual crossing implies $E_{hK} < 1/\alpha$; Markov bounds this probability by $\alpha^{-1}e^{-hKg(\lambda)}$. Maximizing the concave quadratic and solving the inequality for hK finish the proof. □

This maximizes the stated lower certificate, not exact power over all tests. For $g(\lambda) > 0$, the same argument bounds the first annual crossing index T_h by $\mathbb{P}(T_h > k) \le \min\{1, \alpha^{-1}e^{-hkg(\lambda)}\}$. Consequently, with $k_0 = \lceil \log(1/\alpha)/(hg(\lambda)) \rceil$, $\mathbb{E}[hT_h] \le hk_0 + h/(1 - e^{-hg(\lambda)})$. This is not a post-change detection-delay guarantee.

For adaptive bets, strengthen the mean assumption to $\mathbb{E}[Y_t \mid \mathcal{F}_{t-1}] = \delta$ at every date. Let

$$\widehat{\lambda}_k = \Pi_{[0,1/(2\nu)]}\left(\frac{[\widehat{\delta}_{k-1}]_+}{2\nu^2}\right), \qquad \widehat{\delta}_{k-1} = \frac{\sum_{t \le h(k-1)} Y_t}{h(k-1)}, \quad k \ge 2. \tag{15}$$

Set $\lambda_1 = \delta_0/(2\nu^2)$ using a predeclared $0 < \delta_0 \le \nu$. Define

$$R_K = h\nu^2 \sum_{k=1}^{K} (\widehat{\lambda}_k - \lambda^*)^2, \qquad C_K = h\nu^2(\lambda_1 - \lambda^*)^2 + \frac{1}{4}\sum_{k=1}^{K-1} \frac{1}{k}. \tag{16}$$

Theorem 5 (T2b–c: estimation cost and adaptive power)**.** *Under the constant-conditional-mean assumption,*

$$\mathbb{E}R_K \le C_K, \qquad \mathbb{E}\log E_{hK} \ge A_K - C_K, \qquad A_K = \frac{hK\delta^2}{4\nu^2}. \tag{17}$$

For every $r > 0$,

$$\pi_K(\widehat{\lambda}) \ge [1 - C_K/r - \alpha^{-1}e^{-A_K+r}]_+. \tag{18}$$

Taking $r = A_K/2$ yields consistency for fixed $\delta > 0$.

Proof. $Y_t - \delta$ are orthogonal martingale differences, so $\mathbb{E}(\widehat{\delta}_{k-1} - \delta)^2 \le \nu^2/[h(k-1)]$. Jensen gives $\delta \le \nu$; positive-part truncation and projection onto an interval containing the oracle do not increase squared error. Thus $\mathbb{E}(\widehat{\lambda}_k - \lambda^*)^2 \le 1/[4\nu^2 h(k-1)]$. Summing proves the regret bound. The third inequality in (9) gives conditional expected log growth at least $\lambda_t\delta - \lambda_t^2\nu^2$. Complete the square:

$$\lambda\delta - \lambda^2\nu^2 = \frac{\delta^2}{4\nu^2} - \nu^2(\lambda - \lambda^*)^2. \tag{19}$$

This proves the expected growth statement. For power, the inverse-capital argument from T2a shows that

$$Z_T = E_T^{-1} \exp\left\{\sum_{t \le T} (\lambda_t\delta - \lambda_t^2\nu^2)\right\}$$

is a nonnegative supermartingale with expectation at most one. At $T = hK$, $Z_T = E_T^{-1}e^{A_K - R_K}$. Hence

$$\begin{aligned}\mathbb{P}(E_{hK} < 1/\alpha) &\le \mathbb{P}(R_K > r) + \mathbb{P}(Z_{hK} > \alpha e^{A_K - r}) \\ &\le C_K/r + \alpha^{-1}e^{-A_K+r}.\end{aligned}$$

No annual crossing implies the event on the left. Finally $C_K = O(\log K)$ while A_K grows linearly, proving consistency. □

R_K concerns the quadratic growth certificate. It is not asserted to equal the realized log-capital difference from the fixed-bet oracle. In the changing market below, neither a constant effect nor a constant conditional mean is available. T1 still applies when its null is true, but T2b–c cannot be transferred to regime adaptation.

6 Selecting bets for a rejection horizon

The optimizer $\delta/(2\nu^2)$ in T2 maximizes its inverse-capital power certificate. It need not maximize actual crossing probability. Two other objectives are expected log-evidence growth and rejection before a specified deadline. They generally select different bets. The capital in this section is statistical evidence, not investible fund wealth.

6.1 Expected log growth has a different optimizer

For a fixed conditional score law with mean δ and $\mathbb{E}Y^2 \le \nu^2$, write

$$g_{\log}(\lambda) = \mathbb{E}\phi(\lambda Y) - \lambda^2\nu^2/2.$$

Proposition 6 (A strictly concave log-growth objective)**.** *The function $g_{\log}$ is strictly concave on $[0,\infty)$, with*

$$-\frac{5}{4}\nu^2 \le g''_{\log}(\lambda) \le -\frac{3}{4}\nu^2.$$

If $\delta \le 0$, its nonnegative maximizer is zero. If $\delta > 0$, its unconstrained maximizer is the unique solution of

$$\mathbb{E}\left[Y\frac{1+\lambda|Y|}{1+\lambda|Y|+\lambda^2Y^2/2}\right] = \lambda\nu^2, \qquad \frac{4\delta}{5\nu^2} \le \lambda^*_{\log} \le \frac{4\delta}{3\nu^2}. \tag{20}$$

A finite upper cap clips this maximizer.

Proof. Differentiating the specified Catoni function gives

$$\phi'(x) = \frac{1+|x|}{1+|x|+x^2/2}, \qquad \phi''(x) = -\operatorname{sign}(x)\frac{|x|+x^2/2}{(1+|x|+x^2/2)^2}.$$

The second derivative extends continuously by zero at the origin. Setting $u = 1+|x|$ shows that $|\phi''(x)|$ is bounded by the maximum of $2(u^2-1)/(u^2+1)^2$, namely $1/4$. Finite $\mathbb{E}Y^2$ permits dominated differentiation. Thus $g''_{\log} = \mathbb{E}[Y^2\phi''(\lambda Y)] - \nu^2$ has the stated bounds. Its derivative at zero is δ and decreases strictly to a negative value. Integrating the derivative bounds proves (20) and the capped result. □

For small transformed scores, $\lambda^*_{\log}$ is near δ/ν^2, rather than $\delta/(2\nu^2)$. Optimizing a fitted law therefore targets *predicted* log growth; it does not establish optimal growth under an unknown true law. Nor does log-growth optimization solve a finite-deadline power problem. These distinctions are compatible with the existing general betting and log-optimality literature [13].

6.2 Targeting power directly

Fix a working law for future score blocks, a finite action set $\mathcal{A} \subset [0, \lambda_{\max}]$ containing zero, and $b_\alpha = \log(1/\alpha)$. A constant bet within a block of h scores has working log-capital increment

$$Z_\lambda = \sum_{i=1}^{h}\phi(\lambda Y_i) - h\lambda^2\nu^2/2, \qquad Z_\lambda \sim Q_\lambda.$$

Suppose the next block has this conditional working distribution given the chosen action, independently of earlier blocks. Set

$$V_0(\ell) = \mathbf{1}\{\ell \ge b_\alpha\}, \qquad V_k(\ell) = \begin{cases} 1, & \ell \ge b_\alpha, \\ \max\limits_{\lambda\in\mathcal{A}} \int V_{k-1}(\ell+z)Q_\lambda(dz), & \ell < b_\alpha. \end{cases} \tag{21}$$

Proposition 7 (Working-model horizon power: a specialization)**.** *The recursion gives the largest probability, under the specified working model and action class, of crossing at a review within k remaining blocks. A maximizing action chosen from the current log capital and remaining time is predictable. Under the* real *datewise null and real public moment bound, the resulting process retains T1's individual type-I guarantee even when the working power model is misspecified.*

Proof. At zero remaining blocks the claim is the definition of crossing. At an uncrossed state, condition on the next action and increment. The induction hypothesis makes the conditional continuation value at most $V_{k-1}(\ell + z)$; maximizing its integral gives the upper bound. Selecting a maximizer and then the inductively optimal continuation attains it. The finite action set permits measurable tie breaking. Each selected action uses only pre-block information, so T1 applies separately to the real score law. □

This uses standard target reachability [14]. More directly, [22] studies horizon-aware bounded-mean betting, [23] treats rejection-time rewards and Bellman optimization, and [24] optimizes finite-horizon binary-trial e-values. The recursion here is a specialization to block-wise Catoni factors. The computed implementation also discretizes log capital; its numerical maximum is not asserted to be a continuous-control global optimum. A full financial score model would also need the evolving estimation and allocation states. The scalar stationary calculation below does not supply that model.

6.3 A separately frozen controlled comparison

We fix the working score law at $N(.1, 1)$, $\nu^2 = 1.09$, $\alpha = .05$, and the common cap $1/(2\sqrt{1.09})$. Every rule keeps its bet for 100 observations and can reject only at the same 100-observation reviews. The declared deadlines are 400, 800 and 1600 observations. The fixed certificate bet is .0458716 and the exact working-model log-growth bet is .0908020. The horizon rule also uses its current capital and remaining blocks; it is not a single larger constant bet.

Exact Gaussian CDF bins and FFT convolution give block-increment distributions. The primary log-capital spacing is .01 with 129 regular actions; a refinement uses .005, 257 actions and a wider lower boundary. Both include the fixed comparison bets. Their predicted initial powers differ by at most .000396. An initial nearest-grid implementation could round a real subthreshold capital into the absorbing threshold state. The corrected lookup prevents this, and the entire declared random stream is replayed without changing the numerical grids, design alternative, cap or deadlines. Initial outputs remain diagnostics.

The evaluation uses 20,000 independent paths at each actual score mean $0, .1, .2, .3$, all with variance one. All four laws satisfy the common public moment bound. The actual mean determines only the generated scores; bets use the fixed working law, pre-block capital and past scores. Mean zero checks the boundary null; the other two non-design alternatives assess misspecification. The following results concern this controlled score experiment, not additional financial markets or a revised primary fund strategy.

	Null rejection (%)			Design power (%)		
Rule	400	800	1600	400	800	1600
Past-mean certificate	0.11	0.22	0.32	11.41	37.76	78.29
Fixed certificate	0.02	0.19	0.96	4.00	44.94	93.04
Fixed log growth	0.69	1.44	1.98	28.31	66.50	92.44
One-block power	1.20	1.36	1.36	25.53	34.10	36.04
Horizon DP	1.42	1.84	2.25	37.48	72.83	97.07
Refined horizon DP	1.41	1.83	2.26	37.47	72.81	97.13

Table 1: Review-time rejection fractions from 20,000 paths per law. The null has mean zero and the design alternative mean .1, both with variance one. Every rule has the same individual nominal 5% bound, public moment cap and bet/review calendar; observed null rates differ. Each horizon-DP entry uses a separate policy targeting that deadline.

At 400 observations the horizon rule crosses on 7496/20000 paths (37.48%) versus 2281/20000 (11.41%) for the past-mean rule. The paired difference is 26.08 percentage points (Monte Carlo 95% interval [25.47, 26.68]). This compares a common nominal level, not matched actual null-crossing probabilities. At the longest deadline the fixed log-growth rule does not outperform the fixed certificate rule, illustrating that the objectives differ. The repeatedly greedy one-block rule also falls short of the deadline-specific policy.

The refined-grid powers closely agree with the primary implementation, and all six predicted-versus-observed design-power checks pass the declared numerical-plus-Monte-Carlo audit. All pre-block horizon-policy and fixed stakes, every first crossing, and the complete summary are independently reconstructed (12,331,336 scalar checks). Direct multiplicative score replay covers the first sixteen paths under each law; it is not a replay of every score path. Its maximum log-capital discrepancy is 6.17e-14. The stronger alternatives .2 and .3 are retained in the numeric report. No result supplies a power optimum for a changing financial score law or improved fund performance.

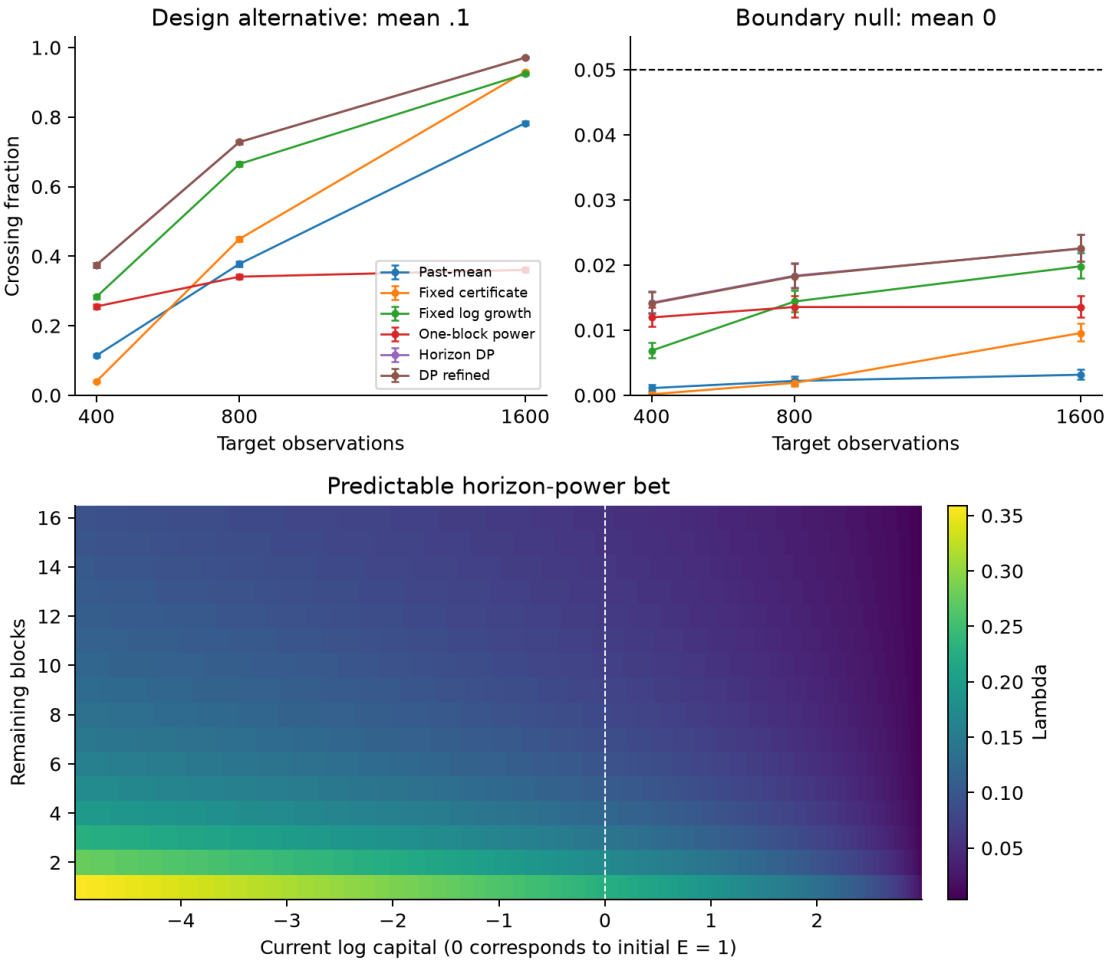


Figure 1: Top: observed design-power and null-crossing fractions with pointwise Wilson 95% intervals. Each DP point uses its own target horizon; the connected points are not one policy monitored at three dates. Bottom: the primary working-model bet as a function of pre-block log capital and remaining blocks. Bets and decisions share the 100-observation calendar.

6.4 A fresh comparison with a deadline-optimized constant

The first experiment compares the Bellman rule with certificate and log-growth bets, which target different objectives. To separate deadline calibration from adaptation to evidence, a new companion experiment chooses the best constant stake on each existing action grid for each deadline. It maximizes working-model probability of crossing at a review, using a constant-action recursion rather than the Bellman maximum at each state. Stake selection and source hashes are frozen before drawing the fresh seed-450001 stream. This is an explicitly subsequent comparison, not an additional arm fitted to the first experiment or extra replications of its paths.

Writing $W_0^\lambda = V_0$, the constant-action recursion is $W_k^\lambda(\ell) = 1$ for $\ell \geq b_\alpha$, and $W_k^\lambda(\ell) = \int W_{k-1}^\lambda(\ell+z)Q_\lambda(dz)$ otherwise. The selected constant maximizes $W_H^\lambda(0)$ over the grid actions, rather than taking a new maximum at every state.

Both grids select the same three constant bets. Their predicted maximum powers differ by at most 0.000465. The fixed-arm simulation holds its stake until crossing or its own deadline, including below the computational lower boundary; absorption there is only a numerical

approximation in its value calculation. All twelve predicted-versus-observed DP/constant design-power checks pass the same declared numerical-plus-Monte-Carlo tolerance.

Deadline	Constant λ	Constant (%)	Horizon DP (%)	Gain (pp)	95% MC interval
400	0.12721	33.10	37.35	4.26	[3.96, 4.55]
800	0.08980	66.02	72.71	6.70	[6.33, 7.06]
1600	0.06361	94.42	96.76	2.34	[2.12, 2.56]

Table 2: Fresh companion comparison at actual mean .1, with 20,000 paired paths, the unchanged nominal individual 5% bound, and the common betting/review calendar. Constants maximize horizon power on the declared working-model action grid. Each deadline uses its own two policies. Gain intervals are pointwise paired normal-approximation Monte Carlo intervals, unadjusted across comparisons; this is not matched-actual-size inference.

At 400 observations the counts are 7470/20000 for horizon DP and 6619/20000 for the optimized constant. The corresponding null fractions are 1.29% and 1.04%. Thus the gain is beyond selecting one better constant at the common nominal level; actual null rejection probabilities remain different. Both numerical resolutions, the unchanged earlier comparators and actual means .2 and .3 are retained in the report. Original and companion powers are reported separately and are not pooled.

The independent companion audit reconstructs every DP/fixed stake and every first crossing (20,497,504 scalar checks). Direct multiplicative score replay covers sixteen paths per law and has maximum log-capital discrepancy 8.88e-14; it is not a replay of every score path. This comparison supplies neither a continuous-action global optimum nor a financial performance claim.

A Gaussian power envelope. The classical Neyman–Pearson bound [26] also applies to early rejection by a fixed deadline. For iid $N(0,\sigma^2)$ versus $N(\delta,\sigma^2)$, $\delta > 0$, any test with actual null rejection probability s by H has power at most

$$B_H(s) = \Phi\Big(\delta\sqrt{H}/\sigma + \Phi^{-1}(s)\Big).$$

Indeed, its rejection event is measurable from the first H scores; their likelihood ratio is increasing in their sum. A terminal upper-tail sum test maximizes power at size s, giving the formula. Independent test randomization does not improve this bound. This oracle fixes the two Gaussian laws: its attaining test need not be valid over the full moment null or under unrestricted monitoring.

At the design alternative, $B_H(.05)$ is 63.88%, 88.17%, 99.07% at the three deadlines. Substituting the observed null fractions of the fresh primary DP rules gives descriptive values 40.94%, 77.12%, 97.60%. These plug-ins do not determine the true null sizes or establish matched-size dominance. The post-hoc report retains both grids, both rule classes and all three alternatives, and gives simultaneous exact-binomial uncertainty limits. It uses the existing audited counts without new paths, threshold changes or policy selection.

7 The economic value of an additional signal

Fix a date and a common previous state b. In this section only, remove the leverage constraint and suppose $H = \gamma M + \eta\Lambda \succ 0$. Writing $a_0 = \mu + \eta\Lambda b$, the objective up to a constant is $a_0'w - w'Hw/2 - c\|w - b\|_1$. Let w_S^* maximize it on support S and let $j \notin S$ with $b_j = 0$. Define

$$q_j = (a_0)_j - H_{jS}w_S^*, \qquad d_j = H_{jj} - H_{jS}H_{SS}^{-1}H_{Sj} > 0. \tag{22}$$

For empty S, use $q_j = (a_0)_j$ and $d_j = H_{jj}$.

Theorem 8 (T3: entry threshold and marginal-value bounds)**.** *The gain* $\Delta_j = V(S\cup\{j\};b) - V(S;b)$ *satisfies*

$$\frac{[|q_j| - c]_+^2}{2H_{jj}} \le \Delta_j \le \frac{[|q_j| - c]_+^2}{2d_j}. \tag{23}$$

In particular, $\Delta_j > 0$ *if and only if* $|q_j| > c$.

Proof. Optimality on S gives $z_S \in \partial\|w_S^* - b_S\|_1$ with $(a_0)_S - H_{SS}w_S^* = cz_S$. Write a perturbation as (u, x) on (S, j). The subgradient inequality and $b_j = 0$ bound its objective improvement by

$$q_j x - c|x| - \tfrac{1}{2}\{u' H_{SS} u + 2xH_{jS}u + H_{jj}x^2\}.$$

Minimizing the quadratic in u gives $u = -H_{SS}^{-1}H_{Sj}x$ and residual curvature d_j. Maximizing $q_j x - c|x| - d_j x^2/2$ gives the upper bound. Holding $u = 0$ makes the actual improvement exactly $q_j x - c|x| - H_{jj}x^2/2$ and gives the lower bound. Positive denominators then prove the entry characterization. □

When $c = 0$, the subgradient inequality is exact and

$$\Delta_j = \frac{\{(a_0)_j - H_{jS}H_{SS}^{-1}(a_0)_S\}^2}{2\{H_{jj} - H_{jS}H_{SS}^{-1}H_{Sj}\}}. \tag{24}$$

With incumbent weights frozen, the new-coordinate optimizer is $\mathrm{sign}(q_j)[|q_j| - c]_+/H_{jj}$. The quantity that matters is residual marginal utility, rather than standalone expected return. The leverage-constrained simulation solves the full convex problem; these unconstrained formulas are not used as constrained shortcuts.

8 From population values to estimated policies

For a common feasible set, let $w^*(S;b)$ maximize J and $\widehat{w}(S;b)$ maximize $\widehat{J}$. Suppose on an event $\mathcal{A}$,

$$\sup_{w,b} |\widehat{J}(w;b) - J(w;b)| \le \varepsilon. \tag{25}$$

This is an assumption about a specified event; it does not itself establish any probabilistic coverage under a changing market.

Theorem 9 (T4: implementation and marginal-value errors)**.** *On* $\mathcal{A}$*, each implementation loss satisfies* $0 \le \ell(S;b) = V(S;b) - J(\widehat{w}(S;b);b) \le 2\varepsilon$*. For two supports at a common state,*

$$|J(\widehat{w}(B;b);b) - J(\widehat{w}(S;b);b) - \{V(B;b) - V(S;b)\}| \le 2\varepsilon. \tag{26}$$

The same bound holds with separate states b_B, b_S *on the corresponding two values, if* (25) *holds uniformly over those states.*

Proof. Insert the estimated objective at the population and estimated maximizers:

$$\begin{aligned} J(w^*;b) - J(\widehat{w};b) = {} & J(w^*;b) - \widehat{J}(w^*;b) + \widehat{J}(w^*;b) - \widehat{J}(\widehat{w};b) \\ & + \widehat{J}(\widehat{w};b) - J(\widehat{w};b) \le 2\varepsilon. \end{aligned}$$

The middle difference is nonpositive. Population optimality gives the lower bound zero. The marginal error is $\ell(S;b) - \ell(B;b)$, the difference of two numbers in $[0, 2\varepsilon]$. Replacing their states changes none of these steps. □

With support-specific uniform error ε_S and estimated solver suboptimality ξ_S, loss is at most $2\varepsilon_S + \xi_S$. The marginal-error bound becomes the maximum of the two loss bounds. For known costs and $\|w\|_2 \le L$,

$$\varepsilon \le L\|\widehat{\mu} - \mu\|_2 + \frac{\gamma L^2}{2}\|\widehat{M} - M\|_{\mathrm{op}}. \tag{27}$$

If Λ is estimated, add $\eta(L + L_b)^2\|\widehat{\Lambda} - \Lambda\|_{\mathrm{op}}/2$.

The same uniform event separates estimation error from the cost of restricting the portfolio. At the fund's actual state b, let $\widehat{G} = \widehat{V}([p]; b) - \widehat{V}(S; b) \ge 0$ and $R = V([p]; b) - J(\widehat{w}(S; b); b)$. Then

$$|R - \widehat{G}| \le 2\varepsilon. \tag{28}$$

Indeed, $|V([p]; b) - \widehat{V}([p]; b)| \le \varepsilon$ and $|J(\widehat{w}(S; b); b) - \widehat{V}(S; b)| \le \varepsilon$. For waiting in cash, $S = \varnothing$ and $b = 0$, the latter difference is exactly zero, so the bound improves to ε. This does not make $\widehat{G}$ an observable confidence bound: the uniform event still needs justification. It shows how support restriction and waiting can dominate even when estimation error is small. The comparator uses the fund's same current state; comparing two entire traded paths requires accounting for their different states as well.

For the link to a common state, assume $\|w\|_2, \|b\|_2, \|b'\|_2 \le L$ and $\|\Lambda\|_{\mathrm{op}} \le \bar{\Lambda}$. Then

$$|C(w, b) - C(w, b')| \le L_C\|b - b'\|_2, \qquad L_C = c\sqrt{p} + 2\eta\bar{\Lambda}L. \tag{29}$$

The maximum over w preserves this Lipschitz bound. Consequently the shadow policy difference d^{state} satisfies

$$|d^{state} - \{V(B; b_S) - V(S; b_S)\}| \le 2\varepsilon + L_C\|b_B - b_S\|_2. \tag{30}$$

This state term cannot be dropped by invoking support monotonicity.

If $\gamma M + \eta\Lambda \succeq mI$, known identical costs and exact optimization yield a refinement. Let $e = \|\widehat{\mu} - \mu\|_2 + \gamma L\|\widehat{M} - M\|_{\mathrm{op}}$. The estimated first-order variational inequality implies that, along every feasible direction, the true negative objective's subgradient is bounded below by the negative gradient error. Strong convexity gives $\ell \le e\|w^* - \widehat{w}\|_2 - m\|w^* - \widehat{w}\|_2^2/2 \le e^2/(2m)$. In fixed-dimensional independent data with finite fourth moments and no persistent moment bias, expected squared moment errors are $O(n^{-1})$. Neither that rate nor consistency follows from expanding estimation under unrestricted drift.

8.1 Optimization horizon and the economic meaning of a move

The distinction between a one-period objective and a policy's holding value matters even with known moments. The following scalar examples specialize standard cost-aware trading calculations [3] to the comparisons being tested; they are not new solutions to the general dynamic trading problem. There is no terminal liquidation charge in either example, matching the simulation's horizon accounting.

Proposition 10 (Parking can overcome myopic inertia)**.** *Let returns be iid with mean zero, variance $\sigma^2 > 0$ and a finite fourth moment. An incumbent allocation $0 < b \le L$ satisfies $\gamma\sigma^2 b \le c$. With known moments, the one-period optimizer on the full scalar support keeps b forever. Comparing immediate parking at zero with that continuing policy, the expected sum of utility differences over H periods is*

$$H\frac{\gamma\sigma^2 b^2}{2} - cb - \frac{\eta}{2}b^2. \tag{31}$$

The first-period gain is negative, while the holding-period gain is positive if $H > 2c/(\gamma\sigma^2 b) + \eta/(\gamma\sigma^2)$.

Proof. At the unchanged target, the smooth marginal is $-\gamma\sigma^2 b$, within the proportional-cost subgradient band. Concavity therefore makes b optimal for the one-period problem. Repeating that rule preserves its state. Parking pays $cb+\eta b^2/2$ once and then has zero utility; continuing at b has no turnover cost and mean utility $-\gamma\sigma^2 b^2/2$ each period. Subtraction gives (31). Its first-period value is at most $-cb/2 - \eta b^2/2 < 0$ under the stated condition. □

For $\gamma = 5$, $c = \eta = 10^{-4}$, $\sigma = .002$, $b = .5$ and $H = 252$, the deployment-day gain is $-.600$ basis points of utility, while the expected holding-period gain is 5.675 basis points. Thus a negative deployment-day diagnostic is compatible with useful parking. This does not contradict support monotonicity: the comparison is with a repeatedly myopic policy. A dynamically optimized full-support policy could itself choose to liquidate. The two shadow states also differ after parking, exactly the distinction retained in (30). Stationarity here supplies the future holding calculation; an episode rejection alone does not supply it. The continuing rule has zero state-matched one-period oracle regret at every date. Parking has positive regret $cb+\eta b^2/2-\gamma\sigma^2 b^2/2$ on deployment and zero thereafter, yet has higher expected cumulative utility when (31) is positive. State-matched regret measures local implementation efficiency; it cannot replace full-path performance or turn the one-period Oracle into an upper bound on cumulative policy value.

Proposition 11 (A profitable horizon trade can be absent from the tested pair)**.** *Let iid returns with a finite fourth moment have $0 < \mu \le c$ and raw second moment $M > 0$. Starting at zero, the known-moment one-period optimizer remains at zero, so adding the scalar support leaves both comparison policies identically in cash. Yet, if $H\mu > c$, a positive constant position held for H periods can have positive expected cumulative utility after its initial entry cost. Within the class of constant nonnegative positions, its optimum is*

$$b_H = \min\left\{L, \frac{(H\mu - c)_+}{H\gamma M + \eta}\right\} > 0. \tag{32}$$

Proof. The one-period gradient at zero is $\mu \le c$, so the proportional-cost band prevents entry and the state remains zero. A constant position has expected cumulative utility $(H\mu - c)b - (H\gamma M + \eta)b^2/2$. Maximizing this concave quadratic on $[0, L]$ yields (32); its positive right derivative at zero ensures a positive value when $H\mu > c$. □

With $\mu = .00005$, return standard deviation .01 and the same γ, c, η, H as above, $M = .0001000025$, $b_H = .099125$ and the constant-position value is 6.195 basis points of cumulative utility. The one-period comparison still has $Y = 0$. The horizon trade belongs to a different policy rule; valid inference for this zero-score pair does not certify that other rule's value. Horizon-aware candidate construction is compatible with T1 when it is predictable and costs are correctly accounted for, but it would change the financial experiment and is not silently substituted in the reported one-period design.

9 An auditable annual selection rule

All methods use initial training $n_0 = h = 252$, expanding sample moments and annual parameter updates. The estimated second moment is

$$\widehat{M}_\rho = (1-\rho)n^{-1}\sum_{s\le n} A_s A_s' + \rho\,\mathrm{diag}\left(n^{-1}\sum_{s\le n} A_s A_s'\right) + \epsilon I. \tag{33}$$

Shrinkage and ridge are common to methods and frozen before evaluation. Starting from $S = \varnothing$, each block opens all additions, all removals, and swaps using the ten excluded signals

with the largest training marginal values. The shortlist is known before the validating returns. Retained comparisons continue their own evidence; a swap leaving the shortlist closes its episode and starts afresh if later reintroduced.

At a block end, choose the eligible addition with largest positive episode mean score. If none exists, consider removals, then swaps. Ties use stable signal identifiers. Eligibility for PARK/REVIVE requires current $L \geq \log 20$. At most one support change occurs per annual revision. The next block charges its actual transition from the fund's previous target. In a 15-year market with one training year, only 14 structural changes are possible, including any terminal decision without a subsequent realized investment period. Recovering all 45 predictive functions is not possible under this calendar.

For $s = |S|$, a pass considers $p + s\min(10, p - s)$ candidate policies plus the shared reference. This is a count of candidate evaluations, not total runtime. Swap10 is a search heuristic without a global-optimality guarantee.

10 One simulated financial market

For $N = 500$ stocks and $D = 30$ observed characteristics, excess returns are

$$R^e_{i,t+1} = \beta_i' F_{t+1} + \sum_{j=1}^{100} \theta_{j,t} f_j(Z_{i,t}) + \epsilon_{i,t+1}. \tag{34}$$

Twenty latent characteristics drive candidate predictors and ten form an independent noise block. Each block follows a stationary Gaussian AR(1) with persistence 0.97, unit marginal variance and correlation 0.6 within five-coordinate groups. Observed characteristics are its componentwise hyperbolic tangent. A design-seeded library contains normalized linear functions, saturating functions and interactions, all bounded by one.

There are 15 persistent, 10 weak, 10 decay and 10 revive functions, together with 40 independent structural-noise functions and 15 correlated proxies. Noise functions use only the independent characteristic block. Each proxy combines 0.9 times a predictor with 0.1 times an independent noise function. All 55 structural nulls have exactly zero coefficient and zero coefficient innovation, while their characteristics continue to vary.

A single symmetric two-state Markov chain starts in state zero and switches with daily probability 1/1008. Stress raises volatility. A first-stress indicator switches decay targets permanently to zero; revive targets are high in normal states and low in stress. Predictive coefficients follow a projected AR(1) with half-life 126 days, bounded symmetric innovations and caps fixed in configuration. No seed is selected for having a particular regime history.

The four factor shocks and stock-specific shocks are independent standardized Laplace variables. Normal-state factor standard deviations are (.009, .004, .004, .004) and stock-specific standard deviation is .012; stress multiplies them by 1.5. Factor means are zero and fixed stock betas are bounded by $(1.5, 1, 1, 1)$ in absolute value. For function values $z_{ij,t}$, the common sleeve weights are

$$P_{ij,t} = \frac{z_{ij,t} - N^{-1}\sum_l z_{lj,t}}{\max\{\sum_l |z_{lj,t} - N^{-1}\sum_k z_{kj,t}|, 10^{-12}\}}, \qquad A_{t+1} = P_t' R^e_{t+1}. \tag{35}$$

Each sleeve is dollar neutral and has gross leverage at most one. Sleeves are not fully factor neutralized. All investible algorithms receive realized sleeve returns and public risk caps; the extension additionally uses current public function values and sleeve exposures. Realized coefficients, regimes and exact conditional moments are confined to the evaluator and oracle.

10.1 A public conditional moment certificate

Suppose both portfolios have conditional return second and fourth moments bounded by m_2, m_4 and costs bounded by C_{max}. Then

$$\mathbb{E}[Y_t^2 \mid \mathcal{F}_{t-1}] \le 12(m_2 + \gamma^2 m_4/4 + C_{max}^2) = \nu^2, \qquad C_{max} = 2cL + 2\eta\bar{\Lambda}L^2. \tag{36}$$

This follows from $U^2 \le 3(r^2 + \gamma^2 r^4/4 + C^2)$ and $(U_B - U_S)^2 \le 2U_B^2 + 2U_S^2$.

Let m bound absolute conditional portfolio mean and v bound variance using the declared coefficient caps, stock beta caps, maximum volatility and gross leverage. Independent standardized Laplace shocks have fourth cumulant three, so one may take $m_2 = m^2 + v$ and $m_4 = m^4 + 6m^2v + 6v^2$. These public quantities do not reveal a realized hidden coefficient or regime. The resulting constant is deliberately conservative. A sample variance is not substituted for it. The ratio of the certificate to true score second moments is reported to diagnose power.

10.2 Benchmarks and evaluation

All Signals optimizes over the entire library. Greedy uses the same search and accepts a positive mean score without a test threshold. Repeated-HAC-p uses the same policy comparisons and repeated annual one-sided normal p-values, with Bartlett long-run variance and bandwidth $\lfloor 4(n/100)^{2/9} \rfloor$. Its repeated testing is uncorrected and finite-sample nominal validity is not claimed. The E-process method uses (10). Oracle uses true current moments with identical costs and constraints, without future shocks. It maximizes one-period conditional utility, not pathwise wealth. A separate oracle matched to each method's actual state supplies implementation regret.

The primary outcome is mean OOS utility. Secondary outcomes are net wealth, conventional and HAC-adjusted Sharpe, drawdown, costs, turnover, inactivity and selected support size. A variance-zero Sharpe is undefined. Net wealth is absorbed at zero on insolvency; failed paths are retained. Structural discovery and conditional economic quality are reported separately. Two consecutive favorable annual economic comparisons define exploratory evaluator events. Those legacy event logs condition on survival of an unchanged comparison; rapid accepted moves can close the comparison before a second block exists. They therefore do not identify comparable detection-delay distributions and are not used as evidence of detection performance. A future delay study must continue the relevant counterfactual policies after their comparisons close. These path-based labels also do not define an ex ante type-I-error calibration experiment.

All methods share each simulated market. Uncertainty about performance uses independent paired seeds, not individual daily observations. Smoke runs test software with reduced dimensions. Pilot and final evaluation use disjoint seed pools with full dimensions; the final design calls for 500 new seeds. Execution status is reported explicitly below.

11 Baseline feasibility and calibration diagnostics

The experiments answer different questions. Table 3 identifies their units of replication and prevents a mechanism study or a software replay from being counted as additional fund-performance evidence. All financial panels use the single stock/characteristic DGP above; the controlled score and scalar Gaussian studies isolate statistical mechanisms.

Panel	Reported replication unit	Question and scope
Baseline search	3 financial markets	Feasibility of the original constant certificate; results immediately below.
First-entry mechanism	20 fresh financial markets	Same fixed-reference scores for all arms; no post-entry fund paths.
Traded-fund pilot	100 financial markets	Complete declared pilot with actual costs, cash waiting and fund paths; every market independently replayed.
Deadline betting	20,000 score paths per law in each of two studies	Original and fresh optimized-constant companion kept separate; no fund strategy.
Estimated-policy cash-PARK	4,096 scalar paths per law	Costless Gaussian reference and adaptive null; no transfer to the financial DGP.
Score calibration	10,000 score paths per law in each diagnostic	Baseline and corrected-tail null checks; not market-level search error calibration.
Final economic assessment	500 registered financial markets	Separate fresh pool; its completed assessment is not reported in this draft.

Table 3: Evidence panels and their inferential units. The complete traded-fund pilot contains all 100 declared markets before the separate final pool. Earlier implementation snapshots are retained without adding their repeated seeds to this count. Counts are not pooled across panels; replaying the same seed with corrected arithmetic adds no independent market. Events within a market are dependent observations.

The three-market baseline below predates the stronger-moment extension and the traded-fund study. It diagnoses the original certificate rather than estimating the final economic contrast. The additional score-calibration experiments use their own laws and independent paths; their rejection fractions do not estimate the overall error of the financial search.

The executed full-dimension pilot contains 3 independent seeds, with $N = 500$, $p = 100$, $h = 252$ and 15 years per seed. These are pilot results, not the planned 500-seed final evaluation. Three separate reduced-dimension smoke seeds exercise the software only.

Method	Utility$\times h$	Net Sharpe	Moves	Inactive
All Signals	0.04685	4.078	0.00	0.000
Oracle	0.05601	4.915	0.00	0.000
Greedy	0.05099	3.951	9.33	0.071
Repeated-HAC-p	0.04335	3.092	4.00	0.071
E-process	0.00000	NA	0.00	1.000

Table 4: Means across executed pilot seeds. Utility is mean daily utility rescaled by h, not an annual certainty equivalent. Inactive paths have undefined Sharpe.

The public bound is $\nu^2 = 0.0132148$. The largest current annual log evidence is 0.006575, compared with $\log 20 = 2.995732$. The small number of seeds is insufficient for a reliable ranking of methods. The conservative certificate and annual opportunity cost must be assessed before any assertion of economic advantage.

Separate controlled statistical diagnostics use 10,000 replications and 1600 observations. Under a predictable heteroskedastic conditional-mean-zero law, the adaptive e-process crosses at rate 0.0036 (Wilson interval [0.0026, 0.0050]). The repeated uncorrected Bartlett-HAC normal test crosses at rate 0.1894. The latter combines finite-sample approximation with repeated testing; these procedures therefore do not have matched overall type-I error. This is an individual-episode calibration experiment, not a financial-market performance result.

Positive constant-mean experiments compare observed power and certificate regret with T2 bounds.

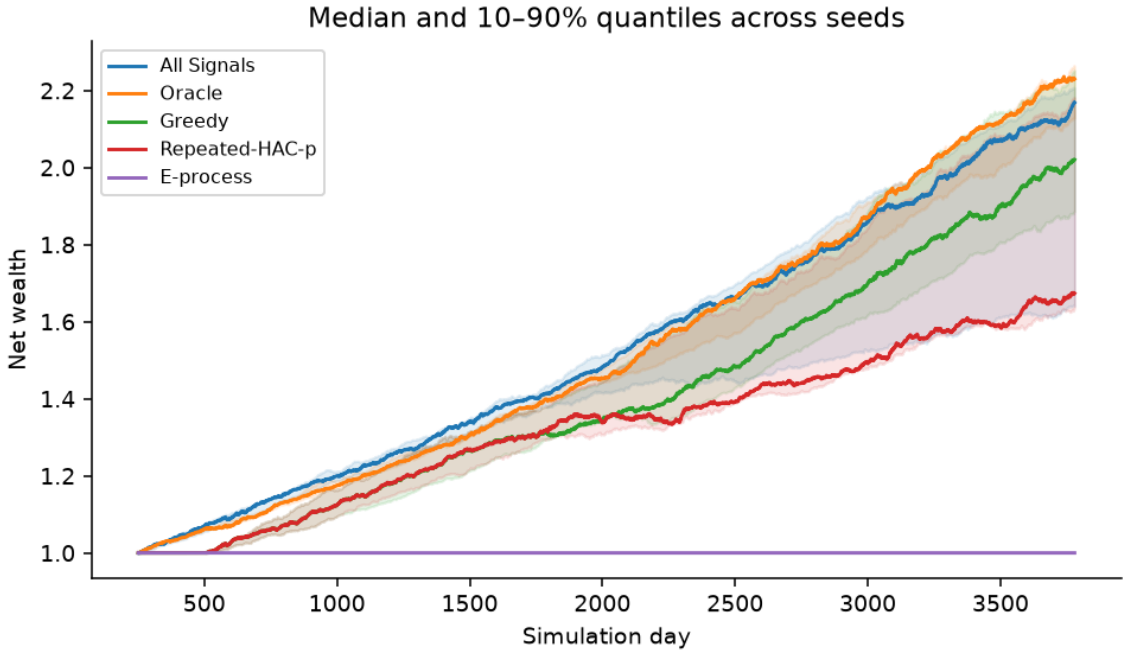


Figure 2: Net wealth median and 10–90 percent quantiles across the executed seeds. Each method uses identical market observations within a seed.

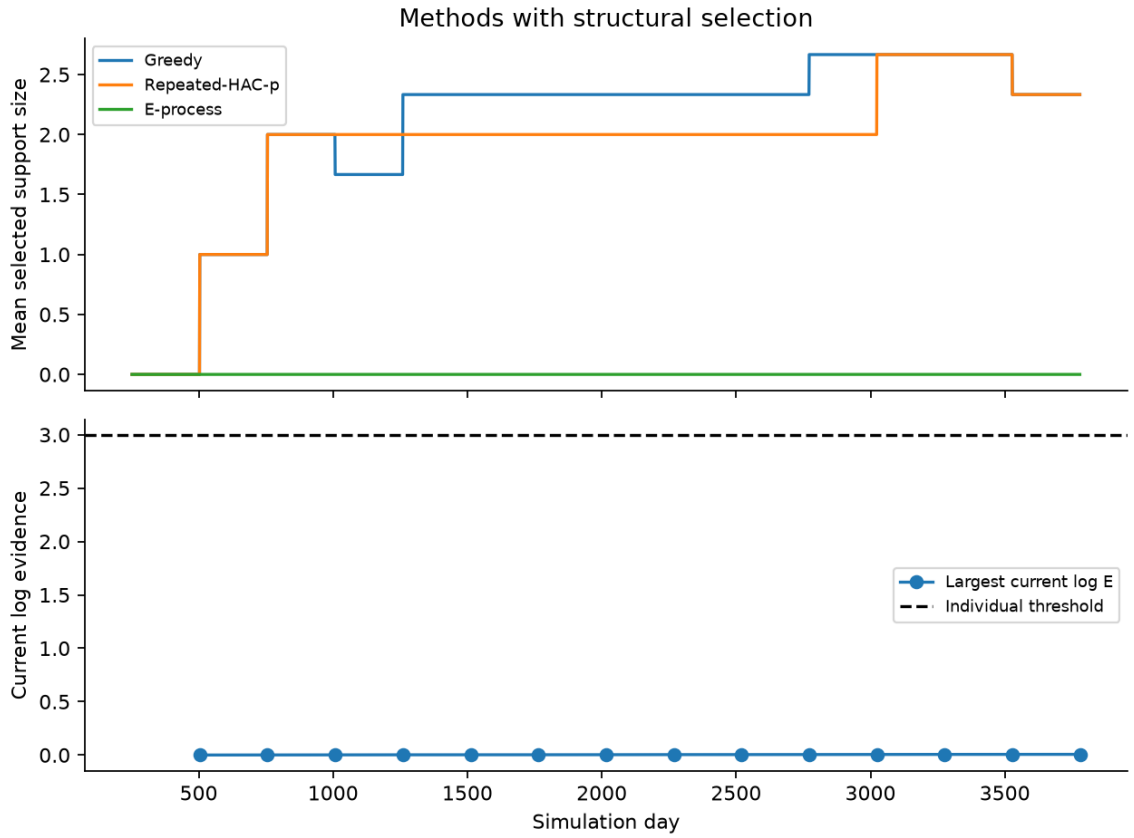


Figure 3: Selected support size for the three search methods and largest current annual E-process evidence. All Signals and Oracle always permit all 100 signals. The threshold is individual; the search has no global 5 percent guarantee.

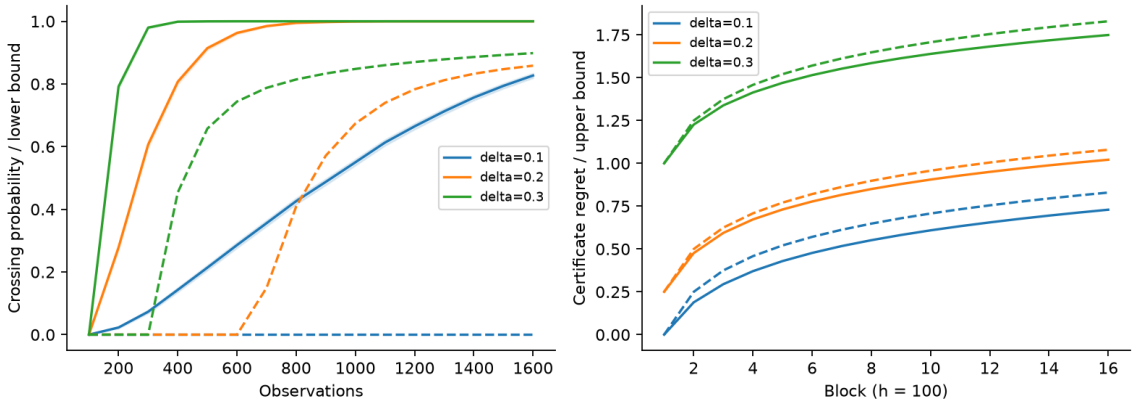


Figure 4: Controlled constant-mean statistical experiments: empirical adaptive power and certificate regret (solid), and theoretical bounds (dashed). These experiments satisfy T2 rather than the regime-changing financial model.

12 Declared extension: observable geometry and corrected tail betting

This section develops an exploratory extension after the baseline pilot. The baseline and its inactive portfolios remain reported unchanged. The extension uses stronger moment information, and its first-entry experiment does not establish the performance of a complete adaptive strategy. Linear betting factors and their mixtures belong to the existing testing-by- betting literature [5]; the contribution here is an explicit policy-moment envelope and a bias-corrected specialization, not a claim to have invented linear betting or mixtures.

12.1 A public envelope without hidden signal labels

Let Z_t^{fun} be the $N \times p$ matrix of observable function values and P_t the observable sleeve holdings before the next stock return R. For a stock allocation $q = P_t w$, conditional return mean is $q' Z_t^{fun} \theta_t$. Replace knowledge of the true support by the public relaxation

$$\Theta = \{\theta \in \mathbb{R}^p : 0 \le \theta_j \le u,\ \textstyle\sum_j \theta_j \le T\}.$$

Here u is the largest coefficient cap and T the sum of group caps; every signal identifier receives the same cap. Knowing these two aggregate bounds does not identify which 55 coefficients are zero.

For a vector a, the support function $h_\Theta(a)$ fills the largest positive coordinates of a at rate u until budget T is exhausted, including a fractional final coordinate. Thus

$$m(q) = \max\{h_\Theta((Z_t^{fun})'q), h_\Theta(-(Z_t^{fun})'q)\}$$

bounds the absolute conditional mean. This is a deterministic linear-program bound, not a fitted mean or an oracle coefficient.

Suppose factor loadings satisfy public componentwise intervals $\beta_{ik} \in [c_k - r_k, c_k + r_k]$, factor scales are at most $\bar\sigma_k$ and idiosyncratic scales at most $\bar\sigma_e$. Independence of the standardized Laplace shocks gives the variance bound

$$v(q) = \sum_k \bar\sigma_k^2 (|c_k \mathbf{1}'q| + r_k \|q\|_1)^2 + \bar\sigma_e^2 \|q\|_2^2. \tag{37}$$

The last term retains cross-sectional diversification rather than replacing the stock L_2 norm by gross leverage. The generator implies intervals $[.5, 1.5]$ for the first beta and $[-1, 1]$ for the other three. The individual realized beta vector remains hidden. Risk scales cover the larger of the normal and stress levels. Dollar neutrality cancels the common center of the first loading interval.

For $x = \mu + h'\zeta$ and independent standardized Laplace shocks, put $\|h\|_2^2 \le v$, $|\mu| \le m$. For even $\ell \le 8$ define

$$H_\ell(m, v) = \sum_{k=0}^{\ell/2} \binom{\ell}{2k} m^{\ell-2k} (2k)! (v/2)^k. \tag{38}$$

Then $\mathbb{E}x^\ell \le H_\ell(m, v)$. To see this, the even Taylor coefficient of $\prod_j (1 - h_j^2 z^2/2)^{-1}$ is bounded by the corresponding coefficient of $(1 - vz^2/2)^{-1}$: each monomial in the former occurs with a multinomial coefficient at least one in the latter. Hence $\mathbb{E}(h'\zeta)^{2k} \le (2k)!(v/2)^k$; binomial expansion supplies (38). No moment-generating assumption is imposed directly on the quadratic utility score.

For two policies, write $d = q_B - q_S$, $s = q_B + q_S$ and predictable $D = C_B - C_S$. Their score is $Y = d'R - \gamma(d'R)(s'R)/2 - D$. Let $H_{\ell,d} = H_\ell(m(d), v(d))$ and similarly for s. Minkowski and conditional Cauchy–Schwarz give

$$V_2 = \left\{ H_{2,d}^{1/2} + \frac{\gamma}{2}(H_{4,d}H_{4,s})^{1/4} + |D| \right\}^2, \tag{39}$$

$$V_4 = \left\{ H_{4,d}^{1/4} + \frac{\gamma}{2}(H_{8,d}H_{8,s})^{1/8} + |D| \right\}^4, \tag{40}$$

as conditional bounds on $\mathbb{E}Y^2$ and $\mathbb{E}Y^4$. For example, $\|(d'R)(s'R)\|_{L_4} \le (\mathbb{E}(d'R)^8\mathbb{E}(s'R)^8)^{1/8}$. These computations may be verified conditional on the simulator's larger filtration; because the bounds use only public pre-return quantities, the tower property gives the same envelopes for the learner's filtration. If targets coincide but prior states differ, $d = 0$ and these formulas give exactly D^2, D^4.

12.2 Bias-corrected lower winsorization

For $p > 1$, $B > 0$ and $x \ge 0$,

$$(x - B)_+ \le \frac{(p-1)^{p-1}}{p^p} \frac{x^p}{B^{p-1}}. \tag{41}$$

The maximum of $(x - B)/x^p$ for $x \ge B$ is attained at $x = pB/(p-1)$, which proves both the inequality and its sharp constant. For predictable envelopes $V_{2,t}, V_{4,t}$ define

$$b_t = \min\left\{ \frac{V_{2,t}}{4B_t}, \frac{27V_{4,t}}{256B_t^3} \right\}, \qquad X_t = \max\{Y_t, -B_t\} - b_t. \tag{42}$$

Only the downside is winsorized; upside may remain unbounded.

Theorem 12 (A corrected-tail e-process for the original utility null)**.** *Suppose $B_t > 0$ is predictable and the two moment envelopes are valid. Under $\mathbb{E}[Y_t \mid \mathcal{F}_{t-1}] \le 0$, the factors*

$$Q_t = 1 + \lambda_t X_t, \qquad 0 \le \lambda_t \le \rho/(B_t + b_t), \quad 0 < \rho < 1,$$

have conditional expectation at most one and satisfy $Q_t \ge 1 - \rho > 0$. Their product is a nonnegative test supermartingale for the original null. Any fixed convex mixture of these products has the same guarantee.

Proof. The increase caused by winsorization is $(-B_t - Y_t)_+$. Apply (41) to $(-Y_t)_+$, using either moment bound, to obtain

$$\mathbb{E}[(-B_t - Y_t)_+ \mid \mathcal{F}_{t-1}] \le b_t.$$

Consequently $\mathbb{E}[X_t \mid \mathcal{F}_{t-1}] \le \mathbb{E}[Y_t \mid \mathcal{F}_{t-1}] \le 0$. Since $X_t \ge -B_t - b_t$, the stated bound on λ_t guarantees positivity. Predictability gives $\mathbb{E}[Q_t \mid \mathcal{F}_{t-1}] \le 1$; multiplication and linearity prove the product and mixture statements. □

Clipping without subtracting the bias can reverse the null mean. For example, a mean-zero distribution with a rare large negative observation becomes positively biased after lower clipping. The correction is therefore part of the test, not an optional robustness adjustment. The fourth-moment route requires eighth return moments, a stronger assumption than baseline T1. A sample fourth moment alone does not supply this certificate.

12.3 A power certificate and the mixture price

Suppose under an alternative that $\mathbb{E}[Y_t \mid \mathcal{F}_{t-1}] \geq \delta_t$. Because winsorization increases the score, $\mathbb{E}X_t \geq \delta_t - b_t$ conditionally. Also $\mathbb{E}[X_t^2 \mid \mathcal{F}_{t-1}] \leq (\sqrt{V_{2,t}} + b_t)^2 =: W_t$. For $z \geq -\rho$,

$$\log(1+z) \geq z - \frac{z^2}{2(1-\rho)}, \qquad \frac{1}{1+z} \leq 1 - z + \frac{z^2}{1-\rho}.$$

The second inequality follows from $(1+z)^{-1} = 1 - z + z^2/(1+z)$; the first follows by differentiation. Thus conditional expected log growth is at least

$$\lambda_t(\delta_t - b_t) - \frac{\lambda_t^2 W_t}{2(1-\rho)}.$$

For a fixed mixture component, put $g_t = \lambda_t(\delta_t - b_t) - \lambda_t^2 W_t/(1-\rho)$. The inverse-capital process compensated by $\exp(\sum g_t)$ is a nonnegative supermartingale. If $\sum_{t \leq hK} g_t \geq G_K$ almost surely, the same endpoint Markov argument as T2a gives the mixture power bound

$$\mathbb{P}\left(\max_{k \leq K} E_{hk}^{mix} \geq 1/\alpha\right) \geq \left[1 - \frac{e^{-G_K}}{\alpha\omega}\right]_+, \tag{43}$$

where ω is that component's fixed mixture weight. This is a lower certificate under explicit alternative assumptions, not an exact power optimum. Pathwise $\log E_t^{mix} \geq \log E_t^{component} + \log \omega$; nine equally weighted components cost at most $\log 9$ relative to one.

12.4 A local information cost of moment-only certification

Conservative moment information can impose a genuine cost even when an alternative generates an entirely stable positive score. The following two-point indistinguishability argument is local to that benign alternative; it is not a minimax rate over all positive-mean distributions. Such rare-event constructions are standard tools in heavy-tail inference; see the broader discussion in [6].

Proposition 13 (A rare-loss lower bound)**.** *Consider any possibly randomized sequential rule with probability at most α of ever rejecting under every i.i.d. law having nonpositive mean, second moment at most v_2 and fourth moment at most v_4. Let P_δ put mass one at $\delta > 0$. For any $0 < \varepsilon < 1$ satisfying*

$$\delta^2 \frac{1-\varepsilon}{\varepsilon} \leq v_2, \qquad \delta^4 \left\{1 - \varepsilon + \frac{(1-\varepsilon)^4}{\varepsilon^3}\right\} \leq v_4, \tag{44}$$

the probability of rejection by observation n under P_δ is at most $\min\{1, \alpha(1-\varepsilon)^{-n}\}$. A necessary condition for power at least $1-\beta$ by n is

$$n \geq \frac{\log\{(1-\beta)/\alpha\}}{-\log(1-\varepsilon)}. \tag{45}$$

Proof. Take an i.i.d. null Q with mass $1-\varepsilon$ at δ and mass ε at $-\delta(1-\varepsilon)/\varepsilon$. Its mean is zero and its moments are exactly the two expressions in (44). With probability $(1-\varepsilon)^n$, its first n observations are all δ. On that event the observed history is identical to the history under P_δ; the rule's independent randomization has the same law. Thus, for rejection event A_n,

$$\alpha \geq Q(A_n) \geq (1-\varepsilon)^n P_\delta(A_n).$$

Rearranging proves both statements. □

For example, if $\varepsilon = \max\{\delta^2/v_2, (2\delta^4/v_4)^{1/3}\} \le 1/2$, both inequalities hold. Indeed the fourth moment is at most $2\delta^4/\varepsilon^3$. With a second-moment restriction alone one may use $\varepsilon = \delta^2/(v_2+\delta^2)$. For small δ, these constructions give local necessary horizons of orders $v_2\delta^{-2}$ and $v_4^{1/3}\delta^{-4/3}$, respectively, up to the logarithmic confidence factor. With both restrictions, the displayed construction has order the smaller of these two expressions.

The corrected-tail construction attains the corresponding local order at a predeclared design effect. For a single known pth-moment bound M_p, put $\kappa_p = (p-1)^{p-1}/p^p$ and choose

$$B = (2\kappa_p M_p/\delta)^{1/(p-1)}, \quad b = \delta/2, \quad \lambda = \rho/(B+b), \quad 0 < \rho < 1.$$

This is a valid test by the clipping proof, and under the deterministic alternative its capital is exactly

$$E_n = \left(1 + \frac{\rho\delta}{2(B+\delta/2)}\right)^n.$$

Consequently its crossing horizon is of order $M_p^{1/(p-1)}\delta^{-p/(p-1)}\log(1/\alpha)$ as $\delta \to 0$. Here the design effect is fixed before testing; this calculation does not license fitting it to the observations used to establish significance. It is an order comparison at a deterministic alternative, not an optimality claim for the nine-component implementation or the changing financial DGP. The two-point null need not belong to the fixed Laplace market: the lower bound applies to guarantees over a full moment class, and can be avoided by legitimately exploiting stronger distributional information.

Corollary 14 (An opportunity cost of certification in a single gate)**.** *Consider the costless single-candidate subcase, with an incumbent of zero utility and deterministic candidate utility $0 < \delta \le 1/(2\gamma)$ per period. A fund remains with the incumbent until the first rejection time τ, then deploys the candidate in the next period. Under the assumptions of the rare-loss proposition, its expected cumulative utility loss over T periods relative to immediate deployment is at least*

$$\delta \sum_{n=0}^{T-1} \left[1 - \alpha(1-\varepsilon)^{-n}\right]_+. \tag{46}$$

Proof. Under the deterministic alternative, lost utility is exactly $\delta \min(\tau, T)$. For an integer-valued stopping time, $\mathbb{E}\min(\tau, T) = \sum_{n=0}^{T-1} \mathbb{P}(\tau > n)$. The preceding proposition bounds each complementary crossing probability, which proves (46). □

This is a single-comparison result in a costless subcase; it neither provides a regret bound for the uncorrected search over 100 signals nor treats the additional trading-state costs of the financial experiment. It isolates a mechanism: evidence waiting can have economic cost even before frictions are introduced. It does not assert that the observed twelve-year waits are necessary in the Laplace market. An explicit quadratic-utility representation takes $r_+ = 2\delta/(1+\sqrt{1-2\gamma\delta})$ and $r_- = -2a/(1+\sqrt{1+2\gamma a})$, with $a = \delta(1-\varepsilon)/\varepsilon$. Then $U(r_+) = \delta$ and $U(r_-) = -a$. This embeds the score null in a fixed ENTRY comparison. It need not preserve additional public return-moment restrictions and does not apply to cash-PARK classes whose economic lower bound excludes the arbitrarily large negative score.

The proposed mechanism experiment fixes scales $c \in \{2, 4, 8\}$, $B_t = cV_{4,t}^{1/4}$, and fractions $\rho \in \{.25, .5, .75\}$ before using new seeds 40001–40020. It retains all nine components in the mixture. Each candidate remains a single-sleeve shadow policy against the empty reference, and all test arms use exactly the same realized scores. Only annual first-entry feasibility is evaluated. Any observations from these new markets are exploratory evidence for this declared extension; they neither replace the old pilot nor constitute a full strategy or final-study result.

12.5 Executed first-entry mechanism study

The executed mechanism study contains 20 fresh markets from seeds 40001–40020, using $N = 500$, $p = 100$ and the 15-year DGP of the baseline. The four arms share every candidate shadow policy and every realized utility score. All 100 comparisons remain under observation to the common horizon, including after crossing. These are fixed-reference mechanism experiments, not twenty new full-strategy backtests.

Arm	Entry markets	Before end	Median year	Max. log E
Catoni–baseline	0/20	0/20	NA	0.008
Catoni–geometry	0/20	0/20	NA	0.555
Tail–uniform	0/20	0/20	NA	0.710
Tail–geometry	19/20	18/20	12.0	4.918

Table 5: Annual eligible first-entry events. "Before end" excludes a first crossing at the terminal revision. Median year is conditional on crossing; censored markets remain in the entry-rate denominator. No actual post-entry fund path is simulated.

The corrected-tail geometric arm has 19 entry markets out of 20; the Wilson 95% interval for this market-level fraction is [0.764, 0.991]. It is a search outcome over 100 candidates, not an individual-test power estimate or a globally controlled rejection probability. One first certification occurs only at the terminal revision; 18 markets have any remaining market observations after certification. Neither geometric Catoni nor uniform-moment corrected-tail betting alone produces an entry in these 20 markets. The combined change improves statistical feasibility but does not remove the substantial cost of waiting.

4 selected first policies use structural proxies whose own coefficient is fixed at zero in the generator. Their conditional shadow gains are positive on the last evaluated date before selection. This illustrates the distinction between structural and economic relevance; it is not a false-discovery calibration result. These conditional gains refer to the shadow state, not a claim that the fund can acquire the shadow positions without paying its actual entry cost.

An additional post-hoc action audit replays all selected histories, refits with then-available observations and computes a hypothetical next-period entry from the fund's actual zero allocation. It values that action with true current model moments after selection, without drawing the next return. 18 of 18 nonterminal entries have positive conditional utility after actual entry costs. The mean is 0.957 basis points of daily utility, compared with 2.457 for the continuing shadow; the actual entry cost is 1.500 basis points. This conditional action valuation is descriptive and supplies no post-entry fund path or new guarantee of current profitability.

Separate predictable skew-null diagnostics use 10,000 independent paths and 2,000 observations. The corrected mixture crosses at rate 0.0068 (Wilson interval [0.0054, 0.0086]). An invalid single bettor ($B_t = V_{4,t}^{1/4}$, fraction .75) without bias correction crosses at rate 0.9966. This is an invalidity demonstration; its configuration differs from the mixture, so it is not a matched power comparison.

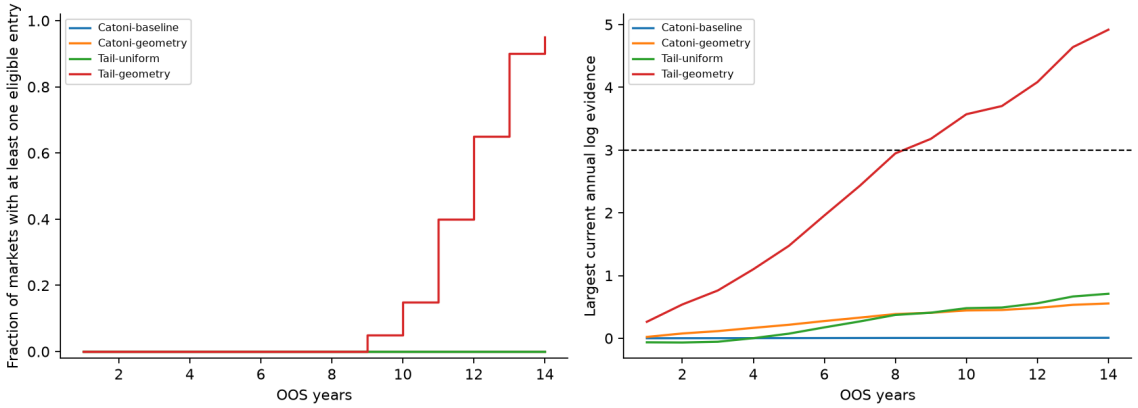


Figure 5: First-entry feasibility across twenty complete fresh markets. Every arm uses the same scores. Curves include terminal test events, with no claim of post-entry trading performance.

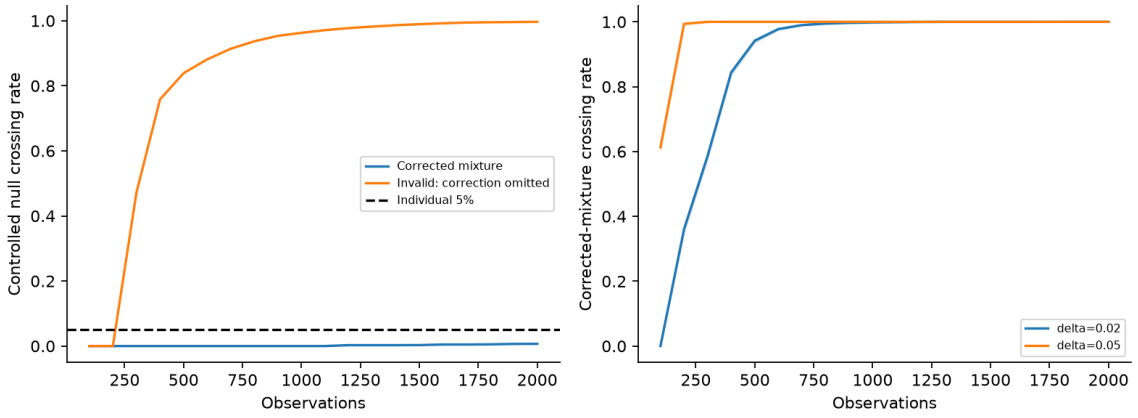


Figure 6: Controlled statistical experiment under a predictable skewed null and two positive-mean alternatives. The invalid negative control is a single uncorrected bettor; it is not a matched power comparison with the corrected mixture. Shading shows Wilson intervals across independent paths.

13 Moment information depends on the direction of the move

The fourth- and eighth-return-moment assumptions above are sufficient for particular score constructions. They are not minimal assumptions for all utility inference. Even a costless fixed-policy comparison shows a directional asymmetry: ENTRY utility is bounded above, whereas cash-PARK utility gain is bounded below. This section concerns explicit iid moment classes; the fixed Laplace financial DGP imposes additional restrictions.

13.1 An exact ENTRY frontier with a second-return-moment cap

Fix $\gamma > 0$ and a known $K_2 > 0$. Let $\mathcal{P}(K_2)$ contain all iid laws of a fixed policy return r with $\mathbb{E}r^2 \le K_2$. The entry score against cash is $U(r) = r - \gamma r^2/2$. Its null is $\mathcal{H}_0 = \{Q \in \mathcal{P}(K_2) : \mathbb{E}_Q U(r) \le 0\}$. The tester may use the entire return sequence and independent randomization.

Theorem 15 (A second-return-moment frontier for uniform ENTRY testing)**.** *Let τ satisfy $\sup_{Q\in\mathcal{H}_0} Q(\tau<\infty)\le\alpha$. For any $P\in\mathcal{P}(K_2)$ with $\mu=\mathbb{E}_P r\le\gamma K_2/2$,*

$$P(\tau<\infty)\le\alpha, \tag{47}$$

including when $\mathbb{E}_P U(r)>0$. Conversely, there is a single valid test, using only K_2 and γ, with eventual power one under every iid $P\in\mathcal{P}(K_2)$ satisfying $\mu>\gamma K_2/2$. The impossibility persists if all laws are restricted to have all absolute moments finite, without common higher-moment bounds.

Proof. Put $m_2=\mathbb{E}_P r^2$. For $0<\varepsilon<1$, define the iid mixture

$$a_\varepsilon=-\sqrt{\frac{K_2-(1-\varepsilon)m_2}{\varepsilon}},\qquad Q_\varepsilon=(1-\varepsilon)P+\varepsilon\delta_{a_\varepsilon}. \tag{48}$$

Its second moment is exactly K_2, and

$$\mathbb{E}_{Q_\varepsilon}U(r)=(1-\varepsilon)\mu-\sqrt{\varepsilon\{K_2-(1-\varepsilon)m_2\}}-\gamma K_2/2<0$$

when $\mu\le\gamma K_2/2$. Coupling the mixtures bounds the total variation distance between the first n return histories by $n\varepsilon$. Therefore $P(\tau\le n)\le\alpha+n\varepsilon$. First let $\varepsilon\downarrow 0$ at fixed n, then $n\to\infty$, proving (47). Finite higher moments are preserved by each finite-point mixture.

For the converse, use the conservative statistic

$$Z=r-\gamma K_2/2,\qquad \mathbb{E}Z\le\mathbb{E}U(r),\qquad \mathbb{E}Z^2\le(\sqrt{K_2}+\gamma K_2/2)^2=:\nu_Z^2. \tag{49}$$

The original null implies $\mathbb{E}Z\le 0$, so T1 applied to Z with its public bound is valid. Under the stated iid alternative, Z has constant positive mean. T2's capped past-mean bet, initialized with gain $\nu_Z/2$, gives crossing probability tending to one. The bet uses no alternative-specific mean, proving the assertion. □

This specializes the total-variation null-density argument in [20, Section 2, Corollary 1]. The proof first controls each finite prefix, then sends the horizon to infinity; it requires no continuity of the test. It is not a new general testing-impossibility principle. It identifies the exact testable region for this utility and information contract. For $\gamma=K_2=1$, deterministic return $r=.2$ has utility .18 but lies in the impossible region. A law's moments may be estimable from an infinite iid sequence while finite-error sequential certification remains impossible uniformly over the class. If $\gamma\sqrt{K_2}\ge 2$, the testable region is empty because $\mu\le\sqrt{K_2}$.

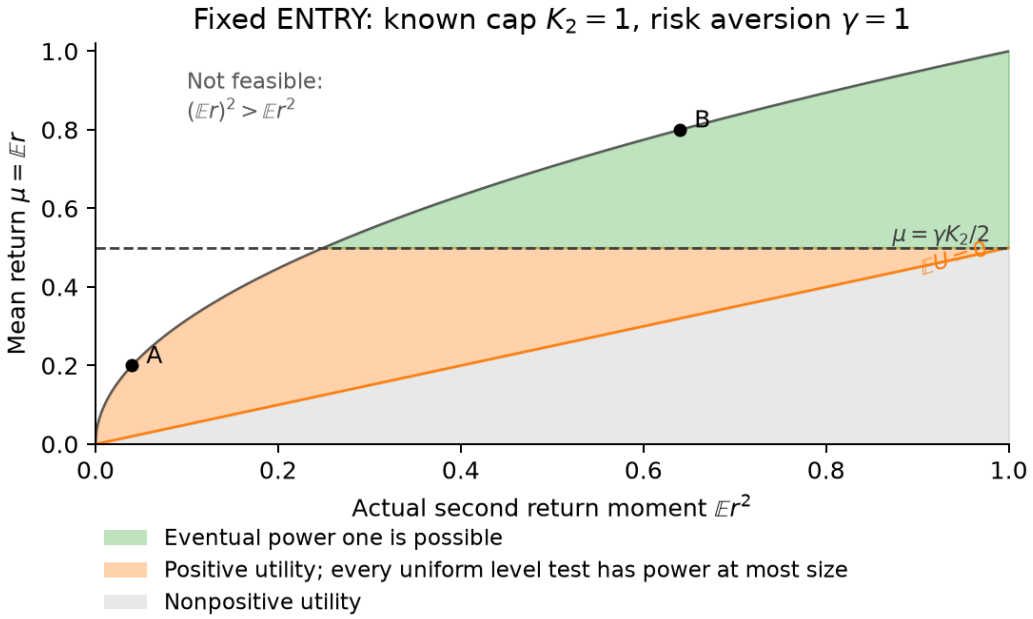


Figure 7: Analytic regions for the fixed-policy iid moment class, not simulated data. Feasibility requires $\mu^2 \le \mathbb{E}r^2 \le K_2$; positive utility requires $\mu > \gamma \mathbb{E}r^2/2$. Uniformly informative ENTRY testing with only the public cap requires the stronger $\mu > \gamma K_2/2$; its boundary belongs to the impossibility region. Point A is deterministic return .2 with utility .18; point B is deterministic return .8 with utility .48. Both are profitable, but only B lies above this moment-class testing frontier.

For a positive-utility law in the impossible region, $m_2 < K_2$. If $m_4 = \mathbb{E}_P r^4 < \infty$, the contaminating null has

$$\mathbb{E}_{Q_\varepsilon} r^4 = (1-\varepsilon)m_4 + \frac{\{K_2 - (1-\varepsilon)m_2\}^2}{\varepsilon} \longrightarrow \infty.$$

Thus a valid numerical higher-moment envelope can rule out these nulls. The second-return-moment surrogate is not merely inefficient in this region: no uniformly valid test of the same class has nontrivial power there. Joint conditional mean/variance tests are established [21]. On the iid class here, our null implies $\mathbb{E}r \le \gamma K_2/2$ and $\operatorname{Var} r \le K_2$. These define a larger null that also contains profitable laws below the frontier. A return-moment test for those bounds is therefore conservative for our utility null; the targets are not equivalent.

13.2 Moments strictly above two can suffice

Proposition 16 (A sufficient higher-moment route)**.** *Consider the iid class with a known $M < \infty$ bounding $\mathbb{E}|r|^{2+\xi}$ for some $\xi > 0$. Set $p = 1 + \xi/2$. A fixed countable mixture of bias-corrected lower-clipped utility betting products is valid for $\mathbb{E}U(r) \le 0$ and has eventual power one at every iid positive-utility law obeying the bound.*

Minkowski and Lyapunov give the required utility moment envelope

$$\mathbb{E}|U(r)|^p \le \left\{ M^{1/(2p)} + \frac{\gamma}{2} M^{1/p} \right\}^p =: A_p. \tag{50}$$

Appendix B gives the complete construction and proof. This existence result uses a countable grid of clipping levels and small stakes, rather than the finite nine-component financial rule.

It asserts no uniform delay or expected-time bound. Fourth/eighth moments are convenient sufficient choices for the reported implementations, not minimal moment orders.

13.3 Cash-PARK uses an economic lower bound

Let $U(r) = r - \gamma r^2/2$, and compare cash with a predictable target $w_t \in [-L, L]$ in a scalar return A_t, with $c = \eta = 0$. The gain is

$$Y_t = -w_t A_t + \frac{\gamma}{2} w_t^2 A_t^2 = \frac{\gamma}{2}(w_t A_t - \gamma^{-1})^2 - \frac{1}{2\gamma}. \tag{51}$$

Consequently the ordinary linear betting process

$$G_n = \prod_{t=1}^{n} (1 + \gamma Y_t/2) \tag{52}$$

has factors at least $3/4$. Under every law with finite conditional score expectations and $\mathbb{E}_{t-1} Y_t \le 0$, it is a test supermartingale, requiring no supplied numerical fourth/eighth-return-moment cap. For iid scores with $\mathbb{E}|Y_t| < \infty$ and $\mathbb{E}Y_t > 0$, a fixed mixture over $\lambda_j = \gamma 2^{-j-1}$, weights 2^{-j-1}, $j \ge 0$, has eventual power one; Appendix B proves this with the small-stake argument. The single stake $\gamma/2$ need not have this universal consistency.

These are specializations of standard safe linear betting and mixtures [2, 5]. The lower bound concerns cash versus this reference; arbitrary reoptimized ENTRY, single-signal deletion and swap scores need not have it. No new financial arm is substituted. The next section treats the dependent scores of an estimated reference and the particular fixed capital (52).

14 Waiting for a vanishing implementation gain

The stationary power bounds above assume a positive effect bounded away from zero. Estimation can instead make an implemented policy's disadvantage shrink over time. This section isolates that mechanism in one costless scalar cash-PARK comparison. It gives a sharp high-confidence exponent for this instance, while the executed experiment below shows extremely low power at a finite horizon. It does not change the financial experiment.

We study the fixed linear capital (52). The universal iid mixture argument above does not by itself cover the dependent score sequence generated by the estimated targets below.

14.1 A stationary estimated-policy instance

Under a law P, let returns, including $n_0 \ge 5$ training observations, be iid $N(0, \sigma^2)$, with $\sigma > 0$. At evaluated date t, compute the historical mean $\bar{A}_{t-1}$ and raw second moment $\widehat{M}_{t-1}$ from all $n_0 + t - 1$ past returns, and set

$$w_t = \mathrm{clip}\left\{ \frac{\bar{A}_{t-1}}{\gamma(\widehat{M}_{t-1} + r)}, [-L, L] \right\}, \quad r \ge 0, \quad L > 0, \quad m = \sigma^2 + r. \tag{53}$$

Use target zero if the denominator vanishes. This makes the rule defined on every history, including laws with degenerate training returns. It is the estimated mean–quadratic one-period optimizer. Although the population optimum is cash, its nonzero fitted targets generate a positive conditional cash-PARK benefit $\delta_t = \gamma \sigma^2 w_t^2/2$. Write $I_n = \sum_{t \le n} w_t^2$ and $C_n = \sum_{t \le n} \delta_t = \gamma \sigma^2 I_n/2$.

Theorem 17 (A sharp cash-PARK confidence exponent)**.** *For (53), under P,*

$$\frac{I_n}{\log n} \longrightarrow \frac{\sigma^2}{\gamma^2 m^2}, \qquad \frac{\log G_n}{\log n} \longrightarrow \kappa = \frac{\sigma^4}{8m^2} > 0 \quad \textit{almost surely.} \tag{54}$$

There is a law Q satisfying the datewise conditional cash-PARK null such that every fixed Q-valid e-process F obeys $\limsup_n \log F_n/\log n \le \kappa$, *$P$-almost surely. For* $\tau_\alpha = \inf\{n : G_n \ge 1/\alpha\}$,

$$\frac{\log \tau_\alpha}{\log(1/\alpha)} \longrightarrow \kappa^{-1}, \qquad \frac{C_{\tau_\alpha}}{\log(1/\alpha)} \longrightarrow \frac{4}{\gamma} \quad \text{almost surely as } \alpha \downarrow 0. \tag{55}$$

For any family of stopping times T_α satisfying $Q(T_\alpha < \infty) \le \alpha$ and any $0 < d < \kappa^{-1}$, $P(T_\alpha \le \alpha^{-d}) \to 0$. Also, for every $\epsilon > 0$,

$$P\left\{\frac{C_{T_\alpha}}{\log(1/\alpha)} < \frac{4}{\gamma} - \epsilon\right\} \longrightarrow 0, \qquad C_\infty = \infty.$$

For the realized cumulative cash advantage $J_n = \sum_{t\le n} Y_t$, $J_n/C_n \to 1$ almost surely. Both stopped-benefit conclusions also hold with J replacing C, with $J_\infty = \infty$. Finite realized sums may be negative. These are limits and probability bounds for cumulative utility, not compounded wealth, expected stopping times or expected stopped benefits.

The proof in Appendix C constructs the null and establishes a finite almost-sure log discrepancy between G and dP/dQ, with uniformly bounded expected absolute discrepancy. Thus sharpness is against this particular estimated-policy alternative and a conditional null permitting dependence. It is not a minimax theorem over investment DGPs. The same null has conditional raw moments at most σ^2 and $3\sigma^4$; the upper bound therefore also applies if these public moment contracts are imposed. With $M_2 = \sigma^2$ and finite $M_4 \ge 3\sigma^4$, geometric Catoni using $v_t^2 = w_t^2(\sigma + \gamma|w_t|\sqrt{M_4}/2)^2$ and fixed bet $\gamma/2$ attains the same κ. Proposition 19 quantifies the effect of a looser M_2 and an adjusted bet. Reviewing only every fixed number of dates preserves the exponent, but a daily threshold crossing need not persist to the next review.

Proposition 18 (A constant certificate can defeat every bet policy)**.** *In the same instance, fix finite $M_4 \ge 3\sigma^4$ and the constant bound $\nu^2 = L^2(\sigma + \gamma L\sqrt{M_4}/2)^2$. For any finite nonnegative predictable bets λ_t, let*

$$\widetilde{E}_n^\lambda = \prod_{t\le n} \exp\{\phi(\lambda_t Y_t) - \lambda_t^2 \nu^2/2\}.$$

There is an explicit finite B_ν, independent of the bet policy, such that for every $u > 0$,

$$\sup_\lambda \; P\left\{\sup_n \widetilde{E}_n^\lambda \ge 1/\alpha\right\} \le \frac{B_\nu}{u} + \alpha e^u. \tag{56}$$

Consequently its maximal ever-rejection probability tends to zero as $\alpha \downarrow 0$, even allowing α-dependent predictable bet rules. For the particular uniform-bound Catoni rule that, after a finite initial block, uses the capped positive past score mean divided by $2\nu^2$ and holds the bet fixed in blocks of a fixed length, log capital converges to a finite real random variable almost surely. In contrast, (52) crosses almost surely at every fixed level. This does not assert zero power at an individual level.

The bound takes a supremum of probabilities over predictable policies, not a pathwise supremum over hindsight choices. The loss combines a shrinking effect and a constant public penalty. The constant-effect experiment of Section 6 has no finite compensator budget, so its power gains are compatible with this obstruction. A fixed positive minimum-effect floor eventually excludes the vanishing gains. Costs, competing closures and changing regimes are absent. Neither result transfers to the full financial PARK/REVIVE rule.

Proposition 19 (Bet selection with a geometry-dependent certificate)**.** *In the same Gaussian instance, fix public $M_2 \ge \sigma^2$ and finite $M_4 \ge 3\sigma^4$, and put*

$$v_t^2 = w_t^2(\sqrt{M_2} + \gamma|w_t|\sqrt{M_4}/2)^2, \qquad q = M_2/\sigma^2 \ge 1.$$

Let $E_n^{\mathrm{geo},\lambda}$ be the Catoni product with penalty $\lambda_t^2 v_t^2/2$. Every fixed constant bet $\lambda \geq 0$ satisfies

$$\frac{\log E_n^{\mathrm{geo},\lambda}}{\log n} \longrightarrow \kappa\left(\frac{4\lambda}{\gamma} - \frac{4q\lambda^2}{\gamma^2}\right) \quad \textit{almost surely.} \tag{57}$$

For each nonnegative predictable bet rule bounded by a finite deterministic constant, the limsup is at most κ/q. The fixed bet $\lambda^* = \gamma/(2q)$ attains this bound, as does the observable rule

$$\widehat{\lambda}_t = \frac{\gamma}{2}\min\left\{\frac{\widehat{M}_{t-1}}{M_2}, 1\right\}. \tag{58}$$

For its first crossing $\widehat{\tau}_\alpha$ of $1/\alpha$,

$$\frac{\log \widehat{\tau}_\alpha}{\log(1/\alpha)} \to \frac{q}{\kappa}, \qquad \frac{C_{\widehat{\tau}_\alpha}}{\log(1/\alpha)} \to \frac{4q}{\gamma}, \qquad \frac{J_{\widehat{\tau}_\alpha}}{\log(1/\alpha)} \to \frac{4q}{\gamma} \quad \textit{almost surely.}$$

The fixed optimizer uses the alternative's σ^2; the predictable rule uses only past raw second moments and the public cap. It changes the bet, not the certificate. A tighter valid cap, including one implied by M_4, changes this comparison. With the unadjusted bet $\gamma/2$, the rate is instead $\kappa(2-q)$ and is negative for $q > 2$. Adjusting the bet restores positive growth but leaves a factor q in the attained waiting coefficient. The optimality statement concerns each fixed bounded predictable rule and the asymptotic log rate; it is not finite-horizon power optimality or a bound for all level-dependent test families. The frozen scalar and financial experiments are unchanged. No expected stopped time or expected stopped benefit is inferred.

Estimation noise in nested forecasts and logarithmic prequential estimation loss are established phenomena [15, 16, 17]. Predictable heteroscedastic Catoni certificates already appear in [1, Section 9]; iid Catoni confidence-sequence width lower bounds are also available [25]. General e-process log-optimality and rejection-time results already exist [13]. Here the score factors are dependent through the fitted policy, the admissible null is adaptive, and the information scale is $\log n$. The explicit cash comparison supplies the exponent and coefficients for conditional and realized benefits. Proposition 18 bounds ever-crossing probabilities over predictable policies in that instance. Likelihood comparison, quadratic strong laws, compensation and Ville's inequality are established proof tools.

14.2 Executed finite-horizon diagnostic

The separately frozen scalar experiment uses 4,096 paths under each of two laws, 65,536 evaluated observations and 32 training observations. It fixes $\gamma = \sigma = 1$, $L = .5$, $r = 0$, and public bounds $M_2 = 1$, $M_4 = 6$. The alternative is the iid Gaussian law of Theorem 17; the boundary null is (61). All four rules see the same return path within each law. The linear and two fixed Catoni stakes are $\gamma/2$; the original uniform-bound past-mean stake updates every 252 observations. No positive minimum-effect filter is applied.

	Crossings / 4096		Alternative final log capital	
Rule	Alternative	Null	Mean	MC SE
Linear cash-PARK	3	0	0.97564	0.01307
Catoni, geometry, fixed	1	0	0.77040	0.01519
Catoni, uniform, fixed	0	0	-5322.34919	0.00782
Catoni, uniform, past mean	0	0	-0.00193	0.00006

Table 6: Controlled scalar cash-PARK diagnostic at individual level 5%. Crossings are over all evaluated dates, not just the last date. The 252-date review counts happen to coincide with these counts at this horizon. These are statistical paths, not financial market replications.

A subsequent summary of the same saved paths gives mean realized cumulative cash advantage $J_n = 3.87129$ (MC SE 0.01583); 14 of 4,096 realized sums are negative. This is a post-hoc fixed-horizon description, not another experiment, a stopped-cost estimate or a verification of the almost-sure limit.

For the same toy configuration, Proposition 18 gives an upper bound of 15.12% on ever rejecting at level 5% for every predictable constant-certificate Catoni bet policy, including level-dependent choices. This is a deterministic upper bound over an unlimited horizon, not an observed crossing fraction or a finite-horizon ranking against the linear rule. The inequality uses an explicit conservative expectation bound for the compensator.

The linear crossing fraction is 0.000732 (Wilson 95% interval [0.000249, 0.002151]); the geometric fraction is 0.000244 ([0.000043, 0.001382]). Three versus one crossing does not establish a finite-sample power ranking. All rules have zero observed alternative crossings at the separately declared 1% and .1% levels. Every null count is zero at all three levels; the Wilson 95% upper bound for each zero count is .000937. Zero observed crossings do not establish zero probability.

The mean integrated squared target is 7.70966, and the mean cumulative conditional cash benefit is 3.85483 in the toy utility units. These are unstopped-horizon means, not expected benefits at rejection or calibrated financial returns. The mean evaluator log likelihood ratio is 0.97827; the linear rule has mean absolute log discrepancy 0.03746. The finite-horizon diagnostics do not establish the asymptotic bounds by themselves.

Independent replay checks 39,888 scalar checkpoint values using direct multiplication of factors and separate Gaussian density calculations. It covers a fixed prefix of 8 paths per law, not all simulated paths; the maximum absolute discrepancy is 8.37e-11. All declared paths and failed-to-cross outcomes are retained. No horizon, bound, bet or seed was retuned after observing low power.

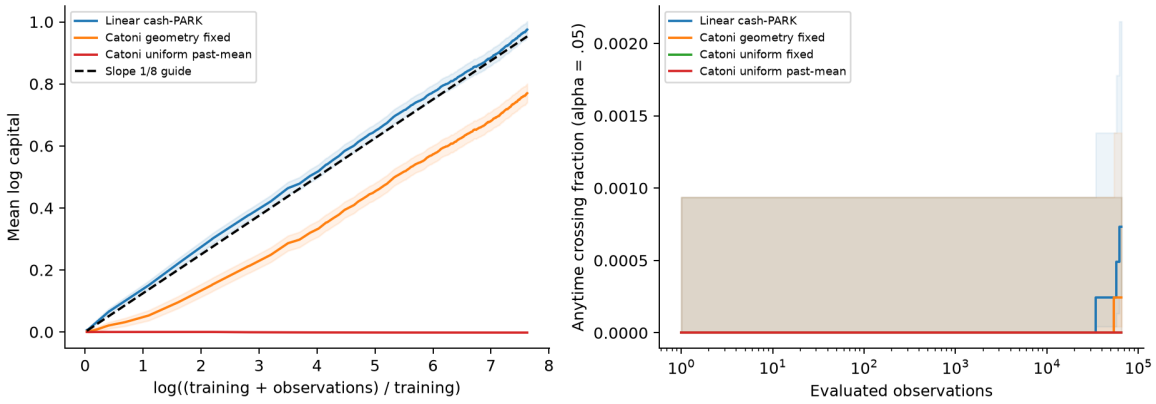


Figure 8: Scalar vanishing-gain diagnostic. Left: mean log capital with pointwise 95% Monte Carlo intervals; the uniform fixed-bet rule is omitted from this panel because its large negative values obscure the other series. The 1/8 line is a slope guide, not an exact finite-horizon formula. Right: all-date crossing fractions at level 5% with pointwise Wilson intervals. Bands are not simultaneous over time.

Eventual crossing and a sharp confidence exponent can coexist with very low finite-horizon power. The experiment supplies no financial performance improvement or adaptation theorem for changing regimes.

15 Several estimated signals: dimension and the waiting coefficient

The scalar calculation extends to a constrained portfolio that estimates a whole second-moment matrix. This extension remains costless and stationary; it is not a result for the changing financial DGP. Dimension changes the confidence-time exponent, while the attained cumulative-utility coefficient remains $4/\gamma$.

Fix a finite number p of signals. Under P, training and subsequent returns are iid $N_p(0,\Sigma)$, with $0 \neq \Sigma \succeq 0$; singular covariance is allowed. At date t, use all $m_t = n_0 + t - 1$ past observations, $n_0 \geq 1$, and define their mean $\widehat{\mu}_t$ and raw second moment $\widehat{M}_t$. For fixed $\rho \in [0,1]$, $r > 0$ and $L > 0$, set

$$\widehat{H}_t = (1-\rho)\widehat{M}_t + \rho\,\mathrm{diag}(\widehat{M}_t) + rI_p, \qquad w_t = \arg\max_{\|w\|_1 \leq L} \{\widehat{\mu}_t' w - \gamma w' \widehat{H}_t w/2\}.$$

The positive ridge makes this optimizer unique on every history. There are no trading costs in this subcase. Cash minus the fitted portfolio has score

$$Y_t = -w_t' A_t + \frac{\gamma}{2}(w_t' A_t)^2, \quad v_t = w_t' \Sigma w_t, \quad C_n = \frac{\gamma}{2}\sum_{t\leq n} v_t, \quad J_n = \sum_{t\leq n} Y_t.$$

Use the same safe linear capital $G_n = \prod_{t\leq n}(1+\gamma Y_t/2)$. It needs no supplied covariance matrix for its validity. Put

$$H = (1-\rho)\Sigma + \rho\,\mathrm{diag}(\Sigma) + rI_p, \qquad \chi = \mathrm{tr}\{(H^{-1}\Sigma)^2\} > 0, \qquad \kappa_p = \chi/8.$$

Theorem 20 (An estimated-matrix cash-PARK comparison)**.** *Under P, almost surely,*

$$\frac{\sum_{t\leq n} v_t}{\log n} \to \frac{\chi}{\gamma^2}, \qquad \frac{\log G_n}{\log n} \to \kappa_p, \qquad \frac{J_n}{C_n} \to 1.$$

There is a conditional cash-PARK null Q, agreeing with P during training, which satisfies

$$\mathbb{E}_Q(A_t A_t' \mid \mathcal{F}_{t-1}) \preceq \Sigma, \qquad \mathbb{E}_Q\{(b'A_t)^4 \mid \mathcal{F}_{t-1}\} \leq 3(b'\Sigma b)^2 \quad \textit{for every } b \in \mathbb{R}^p.$$

Every fixed Q-valid e-process F has $\limsup_n \log F_n/\log n \leq \kappa_p$, P-almost surely. For $\tau_\alpha = \inf\{n : G_n \geq 1/\alpha\}$,

$$\frac{\log \tau_\alpha}{\log(1/\alpha)} \to \frac{8}{\chi}, \qquad \frac{C_{\tau_\alpha}}{\log(1/\alpha)} \to \frac{4}{\gamma}, \qquad \frac{J_{\tau_\alpha}}{\log(1/\alpha)} \to \frac{4}{\gamma} \quad \textit{almost surely.}$$

For every level-indexed family satisfying $Q(T_\alpha < \infty) \leq \alpha$, $P(T_\alpha \leq \alpha^{-d}) \to 0$ for each $0 < d < 8/\chi$. Its conditional and realized cumulative-utility coefficients are at least $4/\gamma$ in the same probability sense as Theorem 17, with infinite costs at $T_\alpha = \infty$. No expected stopped-time or stopped-benefit conclusion is asserted.

The proof is in Appendix D. It reduces the conditional experiment to the one-dimensional projection along the fitted portfolio and proves a matrix integrated-estimation law. The conditional null is adaptive: the upper bound does not assert optimality against a null restricted to iid returns. Training size and dimension are fixed in the high-confidence limit. Neither growing dimension nor $r \downarrow 0$ jointly with the confidence level is covered. The exponents do not bound the duration of the initial constrained phase or finite-horizon power. When $\rho = 0$ and $\Sigma = \sigma^2 I_p$, $\chi = p\{\sigma^2/(\sigma^2+r)\}^2$. More fitted noise directions increase the cash disadvantage and its information at the same leading rate. They change the time exponent while canceling from the stopped-utility coefficient. In general χ describes the learner's limiting curvature; with shrinkage it need not equal the covariance rank or be bounded by p. The ridge remains strictly positive in this theorem; the earlier scalar result also covers zero ridge under its stronger training condition.

Corollary 21 (Bet selection with a matrix cap)**.** *In the same instance, let the public return-moment contract use a fixed $K \succeq \Sigma$: conditional raw second moments are bounded by K and directional fourth moments by $3(b'Kb)^2$. Write*

$$s_t = w_t' K w_t, \qquad \nu_t^2 = s_t\{1 + \gamma\sqrt{3s_t}/2\}^2.$$

Define

$$\mathsf{A} = \Sigma^{1/2} H^{-1} \Sigma H^{-1} \Sigma^{1/2}, \quad \mathsf{B} = \Sigma^{1/2} H^{-1} K H^{-1} \Sigma^{1/2}, \quad \chi_K = \operatorname{tr} \mathsf{B},$$

$$\psi = \mathbb{E}_{Z \sim N_p(0, I_p)} \frac{(Z'\mathsf{A}Z)^2}{Z'\mathsf{B}Z}, \qquad \frac{\chi^2}{\chi_K} \le \psi \le \chi.$$

Set the ratio to zero when its denominator is zero. For Catoni capital with penalty $\lambda_t^2 \nu_t^2/2$, every constant bet has limiting log rate

$$\frac{\chi\lambda}{2\gamma} - \frac{\chi_K \lambda^2}{2\gamma^2}.$$

Its optimum is $\lambda_c = \gamma\chi/(2\chi_K)$, with rate $\chi^2/(8\chi_K)$. Each fixed nonnegative predictable rule with a finite deterministic bound has limsup rate at most $\psi/8$. The past-only rule

$$\widehat{\lambda}_t = \frac{\gamma}{2} \min\left\{ \frac{w_t' \widehat{M}_t w_t}{w_t' K w_t}, 1 \right\}, \qquad \widehat{\lambda}_t = 0 \text{ if } w_t' K w_t = 0.$$

attains $\psi/8$ without knowledge of Σ. Its confidence-time exponent is $8/\psi$ and its conditional and realized cumulative-utility coefficients are $4\chi/(\gamma\psi)$, almost surely. The optimal constant instead attains coefficient $4\chi_K/(\gamma\chi)$.

This is an asymptotic criterion within the prescribed geometric certificate and bounded-rule class, not a deadline-power optimizer or an information bound for every test using K. The upper rate is pointwise in the fixed rule, not uniform over level-dependent policy families. Safe linear capital still attains $4/\gamma$ in this instance. If $K = q\Sigma$, then $\psi = \chi/q$: both optima reduce to $\gamma/(2q)$ and coefficient $4q/\gamma$.

Anisotropy creates a separate gain from adapting the stake. For two signals with $\Sigma = \sigma^2 I_2$ and $K = \sigma^2 \operatorname{diag}(k_1, k_2)$, $k_i \ge 1$, put $a = \{\sigma^2/(\sigma^2 + r)\}^2$. Then

$$\chi = 2a, \quad \chi_K = a(k_1 + k_2), \quad \psi = \frac{2a}{\sqrt{k_1 k_2}}.$$

The adaptive coefficient is $4\sqrt{k_1 k_2}/\gamma$; the best constant's is $2(k_1 + k_2)/\gamma$. For $(k_1, k_2) = (1, 9)$ these are $12/\gamma$ and $20/\gamma$, respectively, while safe linear capital still has $4/\gamma$. These are analytic high-confidence coefficients, not observed finite-level costs or a new arm in the financial experiment.

Corollary 22 (The constant-certificate obstruction persists)**.** *Under the moment contract of Corollary 21, put $V = L^2 \|K\|_{\mathrm{op}}$ and $\nu_*^2 = V\{1 + \gamma\sqrt{3V}/2\}^2$. There is a finite B_*, independent of the predictable bet policy, such that the Catoni product with constant penalty $\lambda_t^2 \nu_*^2/2$ satisfies*

$$\sup_{\lambda} P\{\sup_n \widetilde{E}_n^{\lambda} \ge 1/\alpha\} \le B_*/u + \alpha e^u \quad (u > 0).$$

Consequently maximal ever-rejection probability vanishes as $\alpha \downarrow 0$, including level-dependent predictable policies. This asserts no useful finite level numerical ceiling for a small ridge and changes no financial method.

16 What a search-wide guarantee would cost

The experiments above use an individual level of 0.05. A global budget is possible, but it changes the decision rule and its waiting cost. The following is a standard spending argument, included to make that distinction explicit.

Proposition 23 (Episode spending)**.** *Enumerate adaptively opened episodes by $k \geq 1$, assign deterministic budgets $a_k > 0$ with $\sum_k a_k \leq \alpha$, and reject episode k only when its capital reaches $1/a_k$. Fix a data-generating law P and a set I_0 of episode indices whose conditional null and moment assumptions hold P-almost surely throughout their active dates. An unopened episode is never rejected. Then*

$$P\{\textit{some episode in } I_0 \textit{ is rejected}\} \leq \alpha.$$

The claim holds for each valid capital construction under its own assumptions, including adaptive opening, stopping and closing of comparisons.

Proof. Conditional Ville's inequality at the opening time bounds the rejection probability of each valid null episode by a_k. Stopping its capital at closure preserves this bound. Taking expectations and a countable union bound gives $\sum_{k \in I_0} a_k \leq \alpha$. Restricting rejections further by the positive-history filter or annual priority rule can only reduce this event. No independence between episodes is required. □

This controls rejection of valid episode nulls, not selection of a policy with negative gain on the current date. The latter remains outside T1, as the earlier sign-change counterexample shows. In particular, an ex post negative annual gain is not by itself a label for membership in I_0.

For illustration, reserve half the total budget for the initial 100 episodes, so each receives $a_k = \alpha/200$, and allocate the other half summably to all later episodes. The first-entry threshold becomes $200/\alpha = 4000$, or $\log E \geq 8.294$, rather than $\log E \geq 2.996$. This illustration was examined after the mechanism study; it is not a predeclared simulated comparator. The largest annual log capital in that complete fixed-reference study is 4.918. Consequently none of its 20 markets would produce a first entry under this particular spending rule. Until the first entry, the fixed-empty-reference policies and scores are unchanged, so this is an exact replay implication and does not require a counterfactual post-entry fund path. Other allocations can give different results; this example is not an optimal-budget theorem.

In the stationary power certificates, replacing α by a_k increases the required log evidence by $\log(\alpha/a_k)$. The rare-loss necessary horizon similarly replaces $\log\{(1-\beta)/\alpha\}$ by $\log\{(1-\beta)/a_k\}$. These identities explain a further information cost of protecting a search. They do not quantify actual delay in the changing financial DGP or establish that the uncorrected search has global error exactly equal to its nominal individual level.

17 From first certification to a traded fund

The fixed-reference experiment cannot establish fund performance after entry. We therefore freeze a separate full-strategy study, preserving every earlier result. It uses the same financial DGP, estimation windows, turnover/impact parameters, leverage and annual Forward–Backward–Swap10 priority. Its six methods are All Signals, Greedy, repeated HAC, constant-bound Catoni, geometric corrected-tail betting and the one-period Oracle. All four selection methods start empty; each actual fund pays its own costs and its performance includes all time spent waiting in cash. The Oracle's weights are constructed from current moments before drawing the return shock. It receives no future shock or coefficient path.

The tail method retains the nine scale/fraction pairs already declared. Its geometric moment bounds now use the difference and sum of general candidate and reference stock exposures, with their exact predictable cost difference. No true coefficient, regime or conditional gain enters the bets. The common episode-mean floor is 10^{-8} daily utility, ranking ties within 10^{-9} use stable signal identifiers, and the QP tolerance is 10^{-10}. Empty and single-coordinate allocations are computed exactly. These declared numerical conventions distinguish this study from the earlier positive-only baseline; all four new selection methods share them. The pilot pool is 50001–50100; the separate final pool is 60001–60500. Numerical pilot corrections form allocation differences before projections, evaluate the equivalent score as $d(1-\gamma s/2)-D$, and rescale a failed QP objective before checking its original-unit KKT conditions. They preserve economic parameters and retain failed seeds. Complete ledgers for the nineteen markets shared by the two latest implementations, and the failed market's three previously sealed blocks, match byte for byte. These are integrity checks, not new independent replications. The fund results below use all 100 declared v3 pilot markets, each complete and independently replayed. The earlier first-ten v2 snapshot is retained in the repository, without adding its repeated seeds to this sample. The separate final 500-market assessment remains outstanding. Versioned protocols and failure audits retain the full implementation history.

17.1 The final economic estimand

The final analysis was fixed after inspecting the first-ten v2 and first-twenty v3 pilot results, before any final-market output. Comparator priority was therefore chosen using pilot evidence; it was not preregistered before the pilot. Each final market supplies one paired observation. With $h = 252$ and $14h$ out-of-sample dates, define

$$\bar{U}_i^m = \frac{1}{14h}\sum_{t=1}^{14h} U_{i,t}^m, \qquad D_i = h(\bar{U}_i^{\text{Tail}} - \bar{U}_i^{\text{All Signals}}).$$

Here U^m is actual fund quadratic utility including modeled trading costs and all waiting dates. The primary effect is the mean of D_i over all 500 declared final markets. Its two-sided approximate 95% Monte Carlo interval is $\bar{D} \pm t_{499,.975} s_D/\sqrt{500}$, where s_D is the sample standard deviation across paired markets. This is daily utility scaled by h, not annual return or certainty equivalent. It concerns the fixed DGP and signal library, and does not causally decompose the effect into waiting versus different allocation paths. Greedy, HAC, Catoni and Oracle contrasts are secondary and unadjusted. Sample size, reference, direction and seed membership do not change with the economic sign; a numerical failure does not authorize dropping its market. The exploratory hundred-market pilot intervals below do not replace this complete-sample final analysis.

17.2 Economic events that retain early responses

An economic opportunity starts at the end of the first annual block with positive evaluator conditional mean utility difference. It qualifies as stable only if the unchanged support pair, following its own allocation states and the common expanding-fit rule, has a positive mean in the following block. Inception-positive opportunities and subsequent sign changes are distinct. The delay has annual resolution: zero means selection at that first annual review, not instantaneous recognition of an unobserved daily change.

If selection, a competing move or shortlist withdrawal closes a comparison before its second block, evaluator-only copies of both shadow policies continue for that block. One additional evaluator year resolves pending onsets at the main horizon. The continuation's allocations, utilities and true-mean labels are evaluator outputs and never feed the learner or its betting

capital. Ordinary market returns within the main horizon still update the common estimation history; the extra year is excluded from that history and from actual fund returns. Separate optimizer instances prevent evaluator queries from changing learner warm starts. This construction includes fast responses in the same stability denominator as slower responses. Confirmed events terminate at selection, a competing move, shortlist withdrawal, economic sign reversal or the administrative horizon; those causes are retained separately.

Meeting a statistical criterion differs from winning the annual ranking. We therefore report criterion detection and actual selection separately, with ENTRY, PARK, REVIVE and SWAP lifecycle labels. Within-one/three/five-year fractions require the corresponding full calendar follow-up for every onset, including successful onsets. Competing closures are not treated as independent administrative censoring. These are descriptive, method-specific opportunity populations; they are neither common causal risk sets across methods nor labels for T1's conditional null. Sparse or absent PARK/REVIVE events cannot support a claim of successful regime adaptation.

A separate deployment ledger evaluates the next selected action from the actual previous fund allocation. It compares the new and retained-support weights from the same state under next-period true moments, before drawing that period's shock. This evaluator quantity diagnoses deployment quality; the historical test does not certify it. Its horizon is one day: a negative initial gain can coexist with a positive subsequent holding value after the entry cost is amortized. It is not a classification of lifetime trade value; the full fund utility paths address performance over the study horizon. A choice made at the final review is logged as terminal and contributes no executed trade or subsequent fund performance.

17.3 Numerical verification and executed evidence

The evaluator computes general score moments by writing $d = (w_B - w_S)'A$ and $s = (w_B + w_S)'A$, so $Y = d - \gamma ds/2 - D$. For independent standardized Laplace shocks, the joint cumulant of orders (a, b) of their centered parts is zero when $a + b$ is odd and, for even $a + b = 2k \geq 2$, equals

$$\kappa_{a,b} = \frac{(2k)!}{k2^k} \sum_l h_{d,l}^a h_{s,l}^b.$$

The usual moment–cumulant recurrence then gives all terms needed for $\mathbb{E}Y^2$ and $\mathbb{E}Y^4$. Difference exposures are formed before computing moments to avoid subtracting nearly identical large fourth-order terms. Independent polynomial convolution and conditional Monte Carlo check this calculation. Actual code checks the public geometric certificates against these evaluator-only moments; they are not used to choose the certificate.

An independent arithmetic replay loads saved compact policy weights and recreates the market without loading checkpoints or test-process states. It reconstructs fund wealth, utility and costs; candidate/reference scores and costs; predictable certificates and bets; nine-component tail capitals; annual eligibility and priority selections; and counterfactual event means. This verifies numerical consistency, while proof validity, economic utility and global error control remain separate questions.

The reported full-strategy pilot snapshot contains 100 completed independent markets (0 requested snapshot markets remain incomplete). Each has 500 stocks, 100 signals, one training block and 14 out-of-sample blocks of 252 days. Independent arithmetic replay covers 100 of these completed markets. These are pilot observations; the separate 500-market final study is not asserted to be complete. Earlier baseline and fixed-reference markets, and later markets outside the specified snapshot, are not added to this count.

Method	$h\bar{U}$ (MC SE)	Mean wealth	Inactive	Mean moves
All Signals	0.04429 (0.00113)	1.889	0.000	0.00
Greedy	0.05166 (0.00074)	2.083	0.071	6.47
HAC	0.05100 (0.00075)	2.064	0.071	2.87
Catoni	0.00000 (0.00000)	1.000	1.000	0.00
Tail	0.01104 (0.00043)	1.170	0.821	0.99
Oracle	0.05844 (0.00050)	2.282	0.000	0.00

Table 7: Full-strategy pilot means. Utility is daily utility rescaled by $h = 252$, not an annual certainty equivalent. MC SE uses independent markets. Wealth includes waiting in cash and all modeled actual fund costs. Moves include terminal choices, which create no post-choice fund return.

Tail makes 99 choices (ENTRY: 99, PARK: 0, REVIVE: 0, SWAP: 0), of which 99 are executed before the horizon. Its mean inactive fraction is 0.821. Successful initial certification therefore must be assessed alongside the returns foregone during waiting. The lifecycle counts delimit what this pilot can say about parking and revival.

Method	Eligible stable events	Criterion within year	Selected within year
Greedy	7515	6435	368
HAC	5962	3788	199
Catoni	4787	0	0
Tail	4760	0	0

Table 8: Response-independent stable-event cohorts with a full year of calendar follow-up. Each row has its own endogenous reference policies and event population; raw counts do not identify causal cross-method detection performance. Dependent events are not treated as independent replications.

Method	Executed	Positive gain	Mean gain (bp)	Entry cost (bp)
Greedy	620	0.140	-0.712	1.190
HAC	285	0.305	-0.256	1.291
Tail	99	1.000	0.931	1.500

Table 9: Evaluator-only next-period gains from the actual previous allocation, including modeled transition costs. These descriptive outcomes concern executed decisions, exclude terminal certifications, and are not quantities certified by a historical episode rejection.

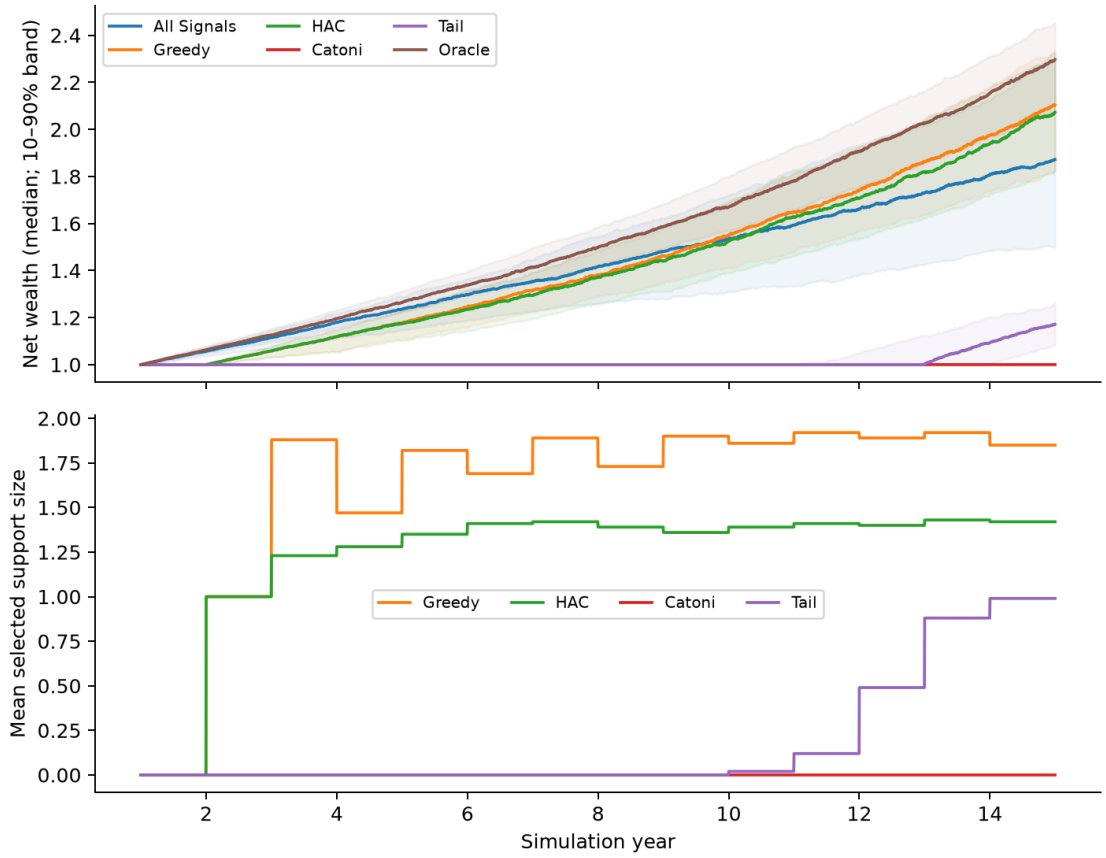


Figure 9: Actual net wealth and selected support size in the completed full-strategy pilot. Wealth bands are the 10th and 90th market percentiles; they are descriptive quantiles, not confidence intervals.

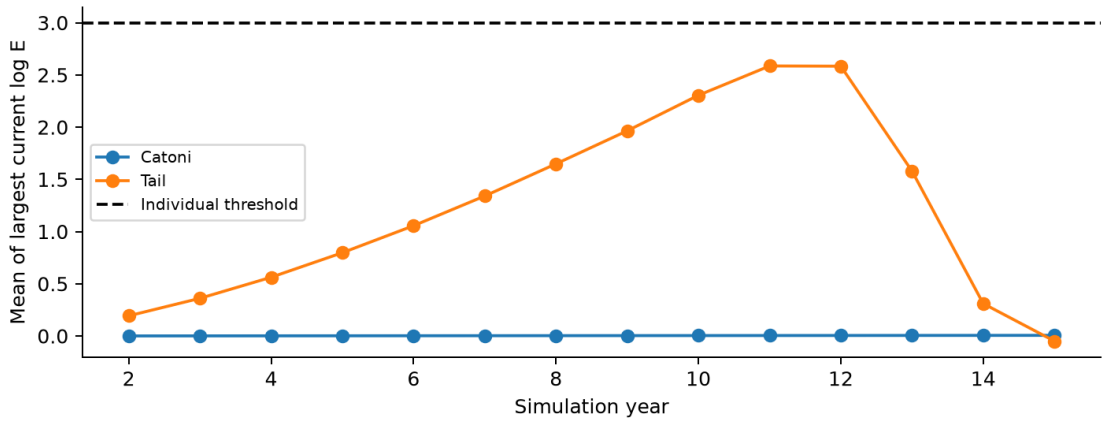


Figure 10: Mean across completed markets of the largest currently open annual log capital. Closed comparisons are not carried forward. The dashed line is the individual threshold; it does not protect the entire search.

Paired Tail-minus-reference differences in $h\bar{U}$ (95% Student MC intervals) are All Signals: -0.03325 [-0.03560, -0.03089]; Greedy: -0.04062 [-0.04187, -0.03938]. These exploratory intervals use independent paired markets and are unadjusted across comparisons; they do not

give guarantees for a new financial law. The reproducibility tables retain every benchmark contrast.

All Signals permits every candidate but does not hold all of them: its mean number of positions exceeding 10^{-6} in absolute sleeve weight is 4.54. This descriptive count is computed after the run and changes no allocation or test. The optimizer already produces a sparse portfolio, so it is a substantive baseline for the value of an additional statistical support filter. No fixed per-signal research cost is charged in this design.

Catoni executes a move in 0 of 100 markets (Wilson 95% interval for the market-level fraction [0.000, 0.037]). A zero observed fraction does not establish zero probability of future entry.

18 Discussion

Moment information and the evolution of the implemented policy jointly determine the scope and cost of validation. The ENTRY frontier separates information about return moments from information about utility scores. With only a cap on the second moment of returns, admissible nulls can remain arbitrarily close to some profitable laws. Known numerical higher moment bounds or distributional restrictions can exclude those nulls. Cash-PARK has an economic lower bound, but estimation can still leave only logarithmically growing information. Its sharp Gaussian confidence exponent and cumulative utility coefficients coexist with extremely low executed finite-horizon power. The common finite compensator budget also bounds every predictable bet policy in the constant-certificate Catoni family, including rules chosen separately for each confidence level. With geometric certification, Proposition 19 instead gives an observable bet attaining the best bounded-rule log rate. If the public second-moment cap is q times the actual variance, its attained cumulative-utility waiting coefficient is multiplied by q, even after adjusting the bet. These statements concern the explicit law classes and policies proved above; they do not make particular gains untestable in the fixed Laplace financial DGP or give expected stopped-cost formulas.

Economic implementation also matters. T3 concerns a direction available to an informed optimizer; T4 and the separate allocation states connect that target to policies that are estimated and traded. A historical episode rejection is distinct from a current-regime profitability claim. An annual calendar limits the number and timing of deployed changes, and no multiplicity guarantee protects the complete reported search.

The baseline certificate prevents entry in the executed pilot. The declared stronger-moment extension permits certification but entails substantial waiting; the separate full-strategy paths include its actual fund costs and time in cash. These pilot observations do not establish practical superiority. The outstanding final simulation must be reported regardless of its economic sign, and a broader prior-art assessment remains necessary for a priority claim. Empirical applications are outside the current scope.

The separate horizon-betting experiment improves rejection probability under its declared stationary design law while retaining individual validity. Its working score model and one-dimensional evidence state do not describe the evolving financial policy pairs. Its power gain therefore does not establish better financial decisions or remove the cost of a conservative public risk envelope, a positive minimum-effect floor or changing regimes.

A Conditional moments used only by the evaluator

Let $x = \mu_x + h_x'\zeta$ and $y = \mu_y + h_y'\zeta$, where the components of ζ are independent, symmetric, variance one and fourth moment six. Write $v_x = h_x'h_x$, $v_y = h_y'h_y$ and $v_{xy} = h_x'h_y$. Then

$$\begin{aligned}
\mathbb{E}x^2 &= \mu_x^2 + v_x, & \mathbb{E}x^3 &= \mu_x^3 + 3\mu_x v_x,\\
\mathbb{E}x^4 &= \mu_x^4 + 6\mu_x^2 v_x + 3v_x^2 + 3\sum_l h_{xl}^4,\\
\mathbb{E}[x^2y] &= \mu_x^2\mu_y + \mu_y v_x + 2\mu_x v_{xy},\\
\mathbb{E}[xy^2] &= \mu_x\mu_y^2 + \mu_x v_y + 2\mu_y v_{xy},\\
\mathbb{E}[x^2y^2] &= \mu_x^2\mu_y^2 + \mu_x^2 v_y + \mu_y^2 v_x + 4\mu_x\mu_y v_{xy}\\
&\quad + v_x v_y + 2v_{xy}^2 + 3\sum_l h_{xl}^2 h_{yl}^2.
\end{aligned}$$

These follow by expanding products and keeping only even multiplicities of each independent centered shock. With $D = C_B - C_S$, $L = x - y$, $Q = x^2 - y^2$,

$$\begin{aligned}
\mathbb{E}Y &= \mathbb{E}L - \gamma\mathbb{E}Q/2 - D,\\
\mathbb{E}Y^2 &= \mathbb{E}L^2 + \gamma^2\mathbb{E}Q^2/4 + D^2 - \gamma\mathbb{E}[LQ] - 2D\mathbb{E}L + \gamma D\mathbb{E}Q,\\
\mathbb{E}L^2 &= (\mu_x - \mu_y)^2 + v_x + v_y - 2v_{xy},\\
\mathbb{E}Q^2 &= \mathbb{E}x^4 + \mathbb{E}y^4 - 2\mathbb{E}[x^2y^2],\\
\mathbb{E}[LQ] &= \mathbb{E}x^3 + \mathbb{E}y^3 - \mathbb{E}[xy^2] - \mathbb{E}[x^2y].
\end{aligned}$$

The implementation verifies these expressions against an independent conditional Monte Carlo sample. Identical policies and states give exactly $Y = 0$, a separate cancellation check.

B Higher-moment and directional consistency arguments

For Proposition 16, conditional versions of (50) hold whenever the stated return-moment contract holds conditionally. Write A_p for that public bound and $\kappa_p = (p-1)^{p-1}/p^p$. Choose $B_i = B_0 2^i$, $i \geq 0$, $B_0 > 0$, and

$$b_i = \kappa_p A_p / B_i^{p-1}, \qquad X_{i,t} = \max\{U(r_t), -B_i\} - b_i, \qquad \lambda_{ij} = \frac{2^{-j-1}}{B_i + b_i}, \quad i, j \geq 0.$$

The sharp tail inequality (41) gives $\mathbb{E}_{t-1}X_{i,t} \leq \mathbb{E}_{t-1}U(r_t)$. Each factor $1 + \lambda_{ij}X_{i,t} \geq 1 - 2^{-j-1} \geq 1/2$ therefore has conditional expectation at most one under the original utility null. With weights $\omega_{ij} = 2^{-i-j-2}$, the sum of the products is a valid e-process by conditional Tonelli. Its finite-time value is finite on every finite return path: $\lambda_{ij} \leq 1/(2B_0)$ and each factor is at most $1 + |U(r_t)|/(2B_0)$, giving a common finite product upper bound.

Under an iid law with $\delta = \mathbb{E}U(r) > 0$, select a fixed i with $b_i < \delta/2$. Then $\mathbb{E}X_i \geq \delta - b_i > 0$. For a fixed integrable X bounded below by $-D$, dominated convergence gives

$$\frac{\mathbb{E}\log(1 + \lambda X)}{\lambda} \longrightarrow \mathbb{E}X \quad \text{as } \lambda \downarrow 0, \quad \lambda D \leq 1/2. \tag{59}$$

For $X \geq 0$ the quotient is at most X; for $-D \leq X < 0$ its absolute value is at most $2|X|$. These bounds prove domination and log integrability. Some j thus has positive expected log factor. The iid strong law makes that component grow to infinity almost surely. Its fixed positive mixture weight proves eventual crossing at every individual level, completing Proposition 16. No finite expectation of the hitting time is needed or inferred.

For the cash-PARK mixture, $Y_t \geq -1/(2\gamma)$ makes every dyadic linear factor positive and of null conditional expectation at most one. The weights sum to one, so the same conditional-Tonelli argument proves validity without a supplied numerical return-moment bound. With iid integrable positive-mean Y, (59) supplies a fixed component with positive log growth and proves eventual power one. This assertion does not apply the iid strong law to the dependent estimated-policy scores of Theorem 17; that theorem has its own proof below.

C Proofs for the vanishing-gain instance

All expectations and almost-sure statements in this appendix, except where marked Q, are under the iid Gaussian alternative P. Training is included in the filtration at evaluated time zero. It has the same distribution under P and the null constructed below, so the initial density ratio is one.

C.1 Integrated estimation error and useful moment bounds

For centered iid observations with finite fourth moment, write $B_i = \sum_{j\le i} A_j$ and $\kappa_4 = \mathbb{E}A_1^4 - 3\sigma^4$. For $i \le j$, independent future increments give

$$\operatorname{Cov}(B_i^2, B_j^2) = 2i^2\sigma^4 + i\kappa_4.$$

Thus $H_n = \sum_{i\le n} B_i^2/i^2$ has $\mathbb{E}H_n = \sigma^2 \log n + O(1)$ and $\operatorname{Var} H_n = O(\log n)$: for the leading covariance term the inner sum over $i \le j$ contributes $O(j)/j^2$, and the fourth-cumulant term contributes $O(\log j)/j^2$. Chebyshev's inequality is summable along $n_k = \lfloor \exp(k^2) \rfloor$. Borel–Cantelli gives $H_{n_k}/\log n_k \to \sigma^2$. Positivity of the summands and $\log n_{k+1}/\log n_k \to 1$ interpolate to all n. The ordinary strong laws make clipping eventually inactive and the risk denominator converge to $m > 0$. Weighted Cesaro and removal of a finite initial prefix yield the first limit in (54).

With k past Gaussian observations, radial/angular independence gives

$$\begin{aligned}
\mathbb{E}(\bar{A}_k/\widehat{M}_k)^2 &= \frac{1}{\sigma^2(k-2)},\\
\mathbb{E}|\bar{A}_k/\widehat{M}_k|^3 &= \frac{2\sqrt{2}\,k^{3/2}}{\sqrt{\pi}\,\sigma^3(k-3)(k-1)(k+1)},\\
\mathbb{E}(\bar{A}_k/\widehat{M}_k)^4 &= \frac{3k}{\sigma^4(k-4)(k-2)(k+2)}.
\end{aligned}$$

For example, rotate the Gaussian vector so its first coordinate is $\sqrt{k}\bar{A}_k/\sigma$, and use its independent length and uniform direction. These formulas apply for $k > 4$. Clipping and $r \ge 0$ decrease absolute targets, hence

$$\mathbb{E}w_t^2 = O(t^{-1}), \qquad \sum_t \mathbb{E}|w_t|^3 < \infty, \qquad \sum_t \mathbb{E}w_t^4 < \infty. \tag{60}$$

We repeatedly use the following martingale normalization. If a centered adapted increment has conditional variance at most Kw_t^2, its sum M_n satisfies $M_n/I_n \to 0$ almost surely. Indeed I_t is predictable and

$$\sum_t \frac{w_t^2}{(1+I_t)^2} \le \int_0^\infty \frac{dx}{(1+x)^2} = 1.$$

For constant K, the martingale of normalized increments is bounded in L^2 and converges; the pathwise Kronecker lemma then gives the assertion. Localization suffices when the predictable variance bound is only almost surely summable.

C.2 A boundary null and the attainable information rate

Let $a_t = \gamma\sigma w_t/2$. Define Q to agree with P during training and thereafter use the sequential Gaussian kernels

$$A_t \mid \mathcal{F}_{t-1} \sim N\!\left(\frac{a_t\sigma}{1+a_t^2}, \frac{\sigma^2}{(1+a_t^2)^2}\right). \tag{61}$$

Here w_t is exactly the same measurable rule on the observed past under both laws. The conditional raw second moment is $\sigma^2/(1+a_t^2)$, so $\mathbb{E}_Q(Y_t \mid \mathcal{F}_{t-1}) = 0$. Its fourth moment is

$$\sigma^4 \frac{3 + 6a_t^2 + a_t^4}{(1+a_t^2)^4} \le 3\sigma^4.$$

These nondegenerate kernels give mutual absolute continuity on every finite horizon without a small-leverage restriction. Put $R_n = dP/dQ|_{\mathcal{F}_n}$ and $Z_t = A_t/\sigma$. Its log increment is

$$\begin{aligned}
\ell_t &= -\log(1+a_t^2) + \tfrac{1}{2}\{[(1+a_t^2)Z_t - a_t]^2 - Z_t^2\}\\
&= -\log(1+a_t^2) + (a_t^2 + a_t^4/2)Z_t^2 - a_t(1+a_t^2)Z_t + a_t^2/2.
\end{aligned} \tag{62}$$

Its conditional mean is $a_t^2/2 + O(a_t^4)$ and its conditional variance is $O(w_t^2)$, uniformly over bounded targets. Martingale normalization, (60) and the integrated-error limit give $\log R_n / \log n \to \kappa$.

The linear factor in (52) is $1 - a_t Z_t + a_t^2 Z_t^2$. There is a finite numerical constant K such that, for all real x,

$$|\log(1 - x + x^2) + x - x^2/2| \le K|x|^3.$$

Taylor expansion proves this near zero; positivity of the quadratic and its logarithmic growth prove it away from zero. Subtracting (62) and expanding $\log(1 + a_t^2)$ yields

$$\log G_n - \log R_n = -\tfrac{1}{2} \sum_{t \le n} a_t^2 (Z_t^2 - 1) + \sum_{t \le n} \rho_t, \qquad \sum_t \mathbb{E}|\rho_t| < \infty. \tag{63}$$

The expectation bound follows from the Gaussian moments, bounded targets, and (60). The displayed martingale is bounded in L^2 since $\sum_t \mathbb{E} a_t^4 < \infty$. Therefore the discrepancy converges almost surely and $\sup_n \mathbb{E}|\log G_n - \log R_n| < \infty$, proving the second limit in (54).

For the geometric Catoni assertion, $|\phi(x) - x| \le K|x|^3$ for all real x, and $v_t^2 - \sigma^2 w_t^2 = O(|w_t|^3 + w_t^4)$. Substituting $x = (\gamma/2) Y_t = -a_t Z_t + a_t^2 Z_t^2$ gives, after summable remainders, the leading log increment $-a_t Z_t + a_t^2 (Z_t^2 - 1/2)$ of (62). Its difference from $\log R_n$ thus has an absolutely summable expected remainder, establishing the same rate.

C.3 Upper bounds, confidence exponents and conditional benefit

For a fixed Q-valid e-process F, any bounded stopping time T satisfies $\mathbb{E}_P(F_T/R_T) = \mathbb{E}_Q F_T \le 1$. Thus F/R is a P-valid e-process. Its path supremum is finite almost surely by Ville's inequality, whence $\limsup_n \log F_n / \log n \le \kappa$. The bound concerns fixed processes; an additional argument is needed for families depending on α.

Since $\log G_n / \log n \to \kappa > 0$, first-crossing times tend to infinity as the threshold grows, and the eventual pathwise bounds $(\kappa - \epsilon) \log n \le \log G_n \le (\kappa + \epsilon) \log n$ sandwich their logarithms. This proves the first limit in (55). The same argument along fixed-spaced review dates proves the review-grid claim without assuming that daily crossings persist.

For an arbitrary level-α family T_α, take $N_\alpha = \lfloor \alpha^{-d} \rfloor$, $d < \kappa^{-1}$, and choose $\eta > 0$ with $\kappa d < 1 - \eta$. Changing measure at the bounded stopped time gives

$$P(T_\alpha \le N_\alpha) \le \alpha^\eta + P\left\{ \max_{t \le N_\alpha} \log R_t > (1 - \eta) \log(1/\alpha) \right\} \longrightarrow 0.$$

The maximum divided by $\log N$ tends to κ almost surely, directly from $\log R_n / \log n \to \kappa$; a finite initial prefix disappears in this normalization. Finally $C_n / \log n \to \sigma^4/(2\gamma m^2)$, so composition at τ_α gives $4/\gamma$. For arbitrary families, monotonicity of C_n and the bound at N_α give every lower coefficient $d\sigma^4/(2\gamma m^2)$; let $d \uparrow \kappa^{-1}$. This proves the probability lower bound for C.

For realized benefits, let $M_n = J_n - C_n$. Gaussian moments give

$$\mathbb{E}_{t-1}(Y_t - \delta_t)^2 = \sigma^2 w_t^2 + \frac{\gamma^2 \sigma^4}{2} w_t^4 \le K w_t^2, \qquad K = \sigma^2 + \gamma^2 \sigma^4 L^2/2.$$

The predictable normalization proved in Section C.1 gives $M_n / I_n \to 0$, and therefore $J_n / C_n \to 1$, almost surely. Composition at τ_α proves the attained realized-benefit coefficient. For arbitrary valid families, the preceding time lower bound implies $T_\alpha \to \infty$ in probability. The eventual pathwise ratio then transfers the probability lower bound from C_{T_α} to J_{T_α}; it does not require monotonicity of J. This completes Theorem 17.

C.4 Proof of Proposition 18

The actual conditional score second moment is $m_{2,t} = \sigma^2 w_t^2 + 3\gamma^2 \sigma^4 w_t^4/4$. The gap

$$d_\nu = \nu^2 - \{\sigma^2 L^2 + 3\gamma^2 \sigma^4 L^4/4\} = \gamma \sigma L^3 \sqrt{M_4} + \gamma^2 L^4 (M_4 - 3\sigma^4)/4 > 0$$

is independent of the bet. The Catoni inequality $e^{\phi(x)} \le 1 + x + x^2/2$ gives

$$\mathbb{E}_{t-1} \exp\{\phi(\lambda_t Y_t) - \lambda_t^2 \nu^2/2\} \le \exp\{\lambda_t \delta_t - d_\nu \lambda_t^2/2\} \le \exp\{\delta_t^2/(2d_\nu)\}.$$

Thus $\widetilde{E}_n^{\lambda} e^{-\mathcal{B}_n}$ is a P-supermartingale for every predictable bet rule, where $\mathcal{B}_n = \sum_{t\le n} \delta_t^2/(2d_\nu)$ is common to all such rules. The fourth-moment bound in (60) gives $\mathbb{E}\mathcal{B}_\infty < \infty$. More explicitly, one may take

$$B_\nu = \frac{3}{8\gamma^2 d_\nu}\left\{\frac{1}{3}\left(\frac{1}{n_0-4}+\frac{1}{n_0-3}\right)+\frac{1}{12}\sum_{j=n_0-2}^{n_0+1}\frac{1}{j}\right\}.$$

Indeed $\mathbb{E}w_t^4 \le 3k/[\gamma^4\sigma^4(k-4)(k-2)(k+2)]$ for $k = n_0+t-1$; partial fractions telescope its sum to the displayed expression. On $\{\mathcal{B}_\infty \le u\}$, crossing $1/\alpha$ requires the compensated supermartingale to cross e^{-u}/α. Ville and Markov therefore give $P(\sup_n \widetilde{E}_n^{\lambda} \ge 1/\alpha) \le \alpha e^u + P(\mathcal{B}_\infty > u) \le \alpha e^u + B_\nu/u$. The bound is uniform in the policy, including policies depending on α; take $u = \frac{1}{2}\log(1/\alpha)$ to obtain its limit. No expected stopped time is used. This specializes standard supermartingale compensation and Ville's inequality to the constant-certificate family.

For the stated plug-in rule, the following sharper pathwise conclusion holds. The score's conditional mean is $\gamma\sigma^2 w_t^2/2$ and its conditional variance is at most Kw_t^2. The normalization just proved gives $\sum_{t\le n} Y_t/I_n \to \gamma\sigma^2/2$, and hence the blockwise past-mean bet is $\lambda_t = O(\log t/t)$ almost surely. Thus $\sum_t \lambda_t^2 < \infty$. Moreover,

$$\mathbb{E}\sum_{t\ge 1}\frac{\log(t+1)}{t}w_t^2 < \infty$$

by (60). The same series is finite almost surely, which implies $\sum_t \lambda_t\delta_t < \infty$. The elementary Catoni bound $|\phi(x)-x| \le x^2/2$ makes the conditional transformed drift absolutely summable, using $\mathbb{E}_{t-1}Y_t^2 \le \nu^2$. The centered transformed increments have conditional variance at most $\lambda_t^2\nu^2$, so their martingale converges by localization. The penalty sum is finite as well. The log capital therefore converges to a finite real limit. Its supremum is finite almost surely, which proves the asserted high-confidence crossing limit.

C.5 Proof of Proposition 19

Fix a predictable rule $0 \le \lambda_t \le \Lambda < \infty$. Gaussian moments and bounded targets give $\mathbb{E}_{t-1}|Y_t|^3 \le K|w_t|^3$. Hence (60) and $|\phi(x)-x| \le K|x|^3$ make $\sum_t |\phi(\lambda_t Y_t) - \lambda_t Y_t|$ finite almost surely. Also $v_t^2 - M_2 w_t^2 = O(|w_t|^3 + w_t^4)$ is summable almost surely. The centered increments $\lambda_t(Y_t - \delta_t)$ have conditional variance at most $K\Lambda^2 w_t^2$, so the normalization in Section C.1 yields

$$\log E_n^{\mathrm{geo},\lambda} = \sum_{t\le n} w_t^2\left(\frac{\gamma\sigma^2\lambda_t}{2} - \frac{M_2\lambda_t^2}{2}\right) + o(I_n) \quad \text{almost surely.} \tag{64}$$

The bracket is maximized at $\lambda^* = \gamma\sigma^2/(2M_2)$, with value $\gamma^2\sigma^4/(8M_2)$. Combining with the limit of $I_n/\log n$ proves the upper bound and (57). The strong law gives $\widehat{M}_{t-1} \to \sigma^2$ and $\widehat{\lambda}_t \to \lambda^*$. Weighted Cesaro in (64) proves its attained rate $\kappa/q > 0$. The crossing-time sandwich used for Theorem 17, followed by $C_n/\log n \to \sigma^4/(2\gamma m^2)$ and $J_n/C_n \to 1$, gives the stated high-confidence coefficients. All limits concern a single fixed policy; no uniformity over level-dependent rules is used.

D Proofs for the estimated-matrix comparison

All limits and moment estimates in this appendix are under P unless marked Q. Dimension, shrinkage, positive ridge, leverage and training size are fixed. Logarithmic averaging of Gaussian and martingale paths has established strong-law formulations [32]. We verify directly the unbounded quadratic forms and ratios needed here; weak convergence of logarithmic empirical measures alone would not justify their moment limits.

D.1 Integrated estimation and moments

Let S_m be partial sums of iid standard p-dimensional Gaussian vectors. For any fixed symmetric $B \succeq 0$ and $i \le j$, Gaussian fourth moments give

$$\mathbb{E}(S_m' B S_m) = m\,\mathrm{tr}\,B, \qquad \mathrm{Cov}(S_i' B S_i, S_j' B S_j) = 2i^2\,\mathrm{tr}(B^2).$$

Therefore $\sum_{m\le n} S_m' B S_m/m^2$ has mean $\operatorname{tr} B \log n + O(1)$ and variance $O(\log n)$. The positive-summand subsequence argument in Appendix C.1 yields

$$\frac{1}{\log n}\sum_{m\le n}\frac{S_m' B S_m}{m^2} \longrightarrow \operatorname{tr} B \quad \text{almost surely.}$$

Apply this with $B = \Sigma^{1/2}H^{-1}\Sigma H^{-1}\Sigma^{1/2}$, whose trace is χ, and with $B = \Sigma$. The ordinary strong laws give $\widehat{H}_t \to H$ and $\widehat{\mu}_t \to 0$. Thus the unconstrained optimizer tends to zero and is eventually feasible for the L_1 constraint. For those dates, the difference between v_t and $\widehat{\mu}_t' H^{-1}\Sigma H^{-1}\widehat{\mu}_t/\gamma^2$ is bounded by $\epsilon_t\|\widehat{\mu}_t\|^2$, with $\epsilon_t \to 0$ almost surely. The second integrated law and weighted Cesaro imply that their accumulated difference divided by $\log n$ vanishes. Removing the fixed training prefix proves $\sum_{t\le n} v_t/\log n \to \chi/\gamma^2$.

The first-order condition along the feasible direction $-w_t$ gives

$$\gamma r\|w_t\|_2^2 \le \gamma w_t'\widehat{H}_t w_t \le \widehat{\mu}_t' w_t, \qquad \|w_t\|_2 \le \frac{\|\widehat{\mu}_t\|_2}{\gamma r}.$$

Since $\widehat{\mu}_t \sim N_p(0, \Sigma/m_t)$, for each fixed positive a, $\mathbb{E}\|w_t\|_2^a = O(m_t^{-a/2})$. In particular

$$\mathbb{E}v_t = O(m_t^{-1}), \qquad \sum_t \mathbb{E}v_t^{3/2} < \infty, \qquad \sum_t \mathbb{E}v_t^2 < \infty,$$

and $v_t \le L^2\|\Sigma\|_{\mathrm{op}}$ deterministically. More explicitly,

$$\mathbb{E}v_t^2 \le \frac{\|\Sigma\|_{\mathrm{op}}^2\{(\operatorname{tr}\Sigma)^2 + 2\operatorname{tr}(\Sigma^2)\}}{\gamma^4 r^4 m_t^2}.$$

These inequalities require no inverse-Wishart moment or growing-dimension approximation. They also apply when Σ is singular.

D.2 A rank-one conditional null

Put $a_t = \gamma\sqrt{v_t}/2$. When $v_t > 0$, define the conditional Gaussian kernel under Q by mean and covariance

$$\mu_{Q,t} = \frac{\gamma\Sigma w_t}{2(1+a_t^2)}, \qquad \Sigma_{Q,t} = \Sigma - \{1-(1+a_t^2)^{-2}\}\frac{\Sigma w_t w_t'\Sigma}{v_t}.$$

Use $N_p(0,\Sigma)$ when $v_t = 0$, and the original training law. In whitened coordinates this changes only the unit direction $e_t = \Sigma^{1/2}w_t/\sqrt{v_t}$: its mean is $a_t/(1+a_t^2)$ and variance $(1+a_t^2)^{-2}$; orthogonal directions retain their original centered Gaussian law. For singular Σ, work on its range. These kernels have the same support and strictly positive densities there at every finite date.

Their raw second-moment matrix is

$$M_{Q,t} = \Sigma - \frac{a_t^2}{1+a_t^2}\frac{\Sigma w_t w_t'\Sigma}{v_t} \preceq \Sigma.$$

Also $w_t'\mu_{Q,t} = \gamma v_t/\{2(1+a_t^2)\}$ and $w_t' M_{Q,t} w_t = v_t/(1+a_t^2)$, proving $\mathbb{E}_Q(Y_t \mid \mathcal{F}_{t-1}) = 0$. For each direction b, a Gaussian variable with mean μ and raw second moment M has fourth moment $3M^2 - 2\mu^4$. Consequently the displayed directional fourth-moment contract follows from $M \le b'\Sigma b$.

Write $R_n = dP/dQ|_{\mathcal{F}_n}$ for these kernels. Under P, $Z_t = w_t'A_t/\sqrt{v_t}$ has conditional standard Gaussian law. Choose a fixed unit whitened direction in the range of Σ when $v_t = 0$; its coefficient a_t is then zero. The likelihood-ratio increment is exactly (62) with this a_t, Z_t, while the linear factor is $1 - a_t Z_t + a_t^2 Z_t^2$. The previous moment estimates provide bounded a_t, $\sum_t \mathbb{E}|a_t|^3 < \infty$ and $\sum_t \mathbb{E}a_t^4 < \infty$. Thus Appendix C's likelihood and summable-discrepancy arguments apply without alteration, with $\sum_t v_t$ as the predictable normalizer. They give $\log R_n/\log n \to \chi/8$, convergence of $\log G_n - \log R_n$ and uniformly bounded expected absolute discrepancy. The score martingale $J_n - C_n$ has conditional variance $v_t + \gamma^2 v_t^2/2$, at most a fixed multiple of v_t. The same martingale normalization gives $J_n/C_n \to 1$.

The fixed-e-process upper bound, the change-of-measure bound for level-indexed stopping rules, and the first-passage inversion in Appendix C now apply with $\kappa = \chi/8$. Since $C_n/\log n \to \chi/(2\gamma)$, their ratio gives the attained and probability-lower-bound coefficient $4/\gamma$. All bounds use the same adaptive null, not a null law selected anew at each confidence level. This proves Theorem 20.

D.3 Geometric and constant certificates

For Corollary 21, the public contracts imply $\mathbb{E}(Y_t^2 \mid \mathcal{F}_{t-1}) \le \nu_t^2$ by Minkowski. Under P, $\nu_t^2 = s_t + O(s_t^{3/2} + s_t^2)$, with summable expected remainder. For each fixed rule with a finite deterministic stake bound, the Catoni expansion and martingale normalization in Appendix C give

$$\log E_n^{\mathrm{geo},\lambda} = \sum_{t \le n} \{\gamma \lambda_t v_t/2 - \lambda_t^2 s_t/2\} + o(\log n) \quad \text{almost surely.}$$

The quadratic strong law also gives $\sum_{t \le n} s_t / \log n \to \chi_K/\gamma^2$.

We need one additional weighted Gaussian law. For standard Gaussian partial sums, put $X_m = S_m/\sqrt{m}$ and $f(x) = (x'\mathsf{A}x)^2/(x'\mathsf{B}x)$, with value zero at a zero denominator. Since $0 \preceq \mathsf{A} \preceq \mathsf{B}$, $0 \le f(x) \le x'\mathsf{A}x$; thus f is even and square integrable. For $i \le j$, X_i, X_j have cross-correlation $\sqrt{i/j}\, I_p$. The centered Gaussian Hermite expansion of an even function has only terms of total degree at least two. Orthogonality therefore gives

$$0 \le \mathrm{Cov}\{f(X_i), f(X_j)\} \le (i/j)\, \mathrm{Var}\{f(Z)\}.$$

Indeed each Hermite term of degree k is multiplied by $(i/j)^{k/2}$. It follows that $\sum_{m \le n} f(X_m)/m$ has mean $\psi \log n + O(1)$ and variance $O(\log n)$. Positivity and the previous subsequence argument prove its ratio to $\log n$ tends to ψ almost surely.

To pass to estimated targets, define $F(w) = (w'\Sigma w)^2/(w'Kw)$ with the same zero convention. Where the denominator is positive,

$$\nabla F(w) = \frac{4(w'\Sigma w)\Sigma w}{w'Kw} - \frac{2(w'\Sigma w)^2 Kw}{(w'Kw)^2}.$$

Because $w'\Sigma w \le w'Kw$, its norm is at most a fixed multiple of $\|w\|_2$. The function extends differentiably with zero gradient on $\ker K$. Hence $|F(w) - F(u)| \le C(\|w\|_2 + \|u\|_2)\|w - u\|_2$. Inverse-curvature convergence, eventual inactivity of the constraint and weighted Cesaro now yield

$$\frac{1}{\log n} \sum_{t \le n} \frac{v_t^2}{s_t} \longrightarrow \frac{\psi}{\gamma^2},$$

where zero-risk terms are zero. This argument works for singular covariance.

Completing the square pointwise gives

$$\gamma \lambda_t v_t/2 - \lambda_t^2 s_t/2 = \frac{\gamma^2 v_t^2}{8 s_t} - \frac{s_t}{2}\left(\lambda_t - \frac{\gamma v_t}{2 s_t}\right)^2$$

when $s_t > 0$. This proves the upper rate $\psi/8$. On the range of K, the strong raw-moment law gives

$$\sup_{w'w>0,\ w'Kw>0} \frac{|w'(\widehat{M}_t - \Sigma)w|}{w'Kw} \to 0 \quad \text{almost surely.}$$

The numerator vanishes on $\ker K$; use its inverse square root on its range. Clipping at one is nonexpansive, so the observable stake differs from $\gamma v_t/(2 s_t)$ by $o(1)$ uniformly in directions. Since $\sum s_t / \log n$ has a finite limit, its accumulated square-completion loss is $o(\log n)$. It attains $\psi/8$. Constant stakes give the stated quadratic rate. Cauchy–Schwarz yields $\chi^2 \le \psi \chi_K$, while $f(Z) \le Z'\mathsf{A}Z$ gives $\psi \le \chi$. First-passage inversion and the utility limits prove the coefficient claims.

For deterministic evaluation of ψ, Gaussian integration and $1/y = \int_0^\infty e^{-ty}\, dt$ give

$$\psi = \int_0^\infty \frac{2\,\mathrm{tr}\{(\mathsf{A}C_t)^2\} + \{\mathrm{tr}(\mathsf{A}C_t)\}^2}{\sqrt{\det(I_p + 2t\mathsf{B})}}\, dt, \qquad C_t = (I_p + 2t\mathsf{B})^{-1}.$$

Tonelli applies to the nonnegative integrand; singular zero directions contribute no numerator. In two isotropic dimensions, separating the Gaussian radius and angle gives the displayed geometric-mean formula.

For Corollary 22, under P,

$$\mathbb{E}(Y_t^2 \mid \mathcal{F}_{t-1}) = v_t + 3\gamma^2 v_t^2/4 \le V + 3\gamma^2 V^2/4, \qquad d_* := \nu_*^2 - V - 3\gamma^2 V^2/4 = \gamma\sqrt{3}\, V^{3/2} > 0.$$

The policy-independent compensator is $\mathcal{B}_n = \sum_{t \le n} \gamma^2 v_t^2/(8d_*)$. Its expected total is at most the finite constant

$$B_* = \frac{\|\Sigma\|_{\mathrm{op}}^2\{(\operatorname{tr}\Sigma)^2 + 2\operatorname{tr}(\Sigma^2)\}}{8\gamma^2 r^4 d_*} \sum_{m=n_0}^{\infty} m^{-2}.$$

Completing the square in the conditional Catoni-factor expectation, then Ville and Markov, gives $B_*/u + \alpha e^u$ exactly as in the proof of Proposition 18. Taking $u = \log(1/\alpha)/2$ proves convergence to zero. The bound may be numerically uninformative for a very small positive ridge; no finite-level efficiency claim is deduced.

E Reproducibility and numerical conventions

Versioned configurations, source hashes, installed versions and seed manifests identify each run. Unfinished runs resume at atomic annual checkpoints without counting a score twice; completed weight, score and event ledgers are retained. The learner is deterministic given its observed history. The market, statistical diagnostics and independent moment checks use distinct random streams. Original input provenance is preserved in the accompanying repository.

The optimizer solves a convex quadratic program with epigraph variables for turnover and gross leverage. It records primal and dual residuals and the duality gap. Exact zero-trade cases have a sufficient subgradient check. Weights, realized score histories and decisions are tested for invariance to future-data perturbations. Stable identifiers govern tie breaking, and the candidate shortlist is tested under a consistent permutation of columns. No old empirical or simulated result enters this manuscript.

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